# Build Data Pipelines with Lakeflow Spark Declarative Pipelines

**Databricks Academy**
October 2025

# Course Learning Objectives

- Understand the core concepts and components of Lakeflow Spark Declarative Pipelines (SDP), including the function and differences between streaming tables, materialized views, and temporary views.
- Identify and configure pipeline settings, such as compute, data assets, trigger modes, and advanced options.
- Develop a functional Spark Declarative Pipeline using the new pipeline editor and SQL-based syntax.
- Incorporate data quality expectations into a pipeline to validate and enforce data integrity.
- Analyze event logs and pipeline metrics to understand the full execution and lifecycle of a Spark Declarative Pipeline.
- Design and implement Change Data Capture (CDC) to a pipeline using AUTO CDC INTO to handle slowly changing dimensions (SCD).

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Agenda

| Course Sections |
| --- |
| • Introduction to Data Engineering in Databricks |
| • Lakeflow Spark Declarative Pipeline Fundamentals |
| • Building Lakeflow Spark Declarative Pipelines |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Course Prerequisites (REQUIRED)

## Fundamental Knowledge of the Databricks Platform

- Databricks Workspaces
- Apache Spark
- Delta Lake and the Medallion Architecture
- Unity Catalog

## Knowledge of Ingesting Raw Data into a Table

- Familiarity with ingesting raw data source files using the **read_files** SQL function
  - CSV
  - JSON
  - TXT
  - Parquet

## Experience Transforming Data with SQL

- Experience in writing **intermediate-level queries** using SQL
- Basic knowledge of **SQL Joins**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Lab Exercise Environment

## Technical Details

- Your lab environment is provided by Vocareum.
- It will open in a new tab.
- It has been configured with the permissions and resources required to accomplish the tasks outlined in the lab exercise.
- Third party cookies must be enabled in your browser for Vocareum's user experience to work properly.
- Make sure to enable pop ups!

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Introduction to Data Engineering in Databricks

**Build Data Pipelines with Lakeflow Spark Declarative Pipelines**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Section Learning Objectives

- Understand the components of Data Engineering with Lakeflow in Databricks with Connect, Spark Declarative Pipelines, and Jobs.
- Explain how Lakeflow Spark Declarative Pipelines incrementally processes data using Streaming Tables and Materialized Views in batch or streaming Lakeflow Jobs.
- Demonstrate how to create a Lakeflow Spark Declarative Pipelines project in a Databricks Workspace.

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Agenda

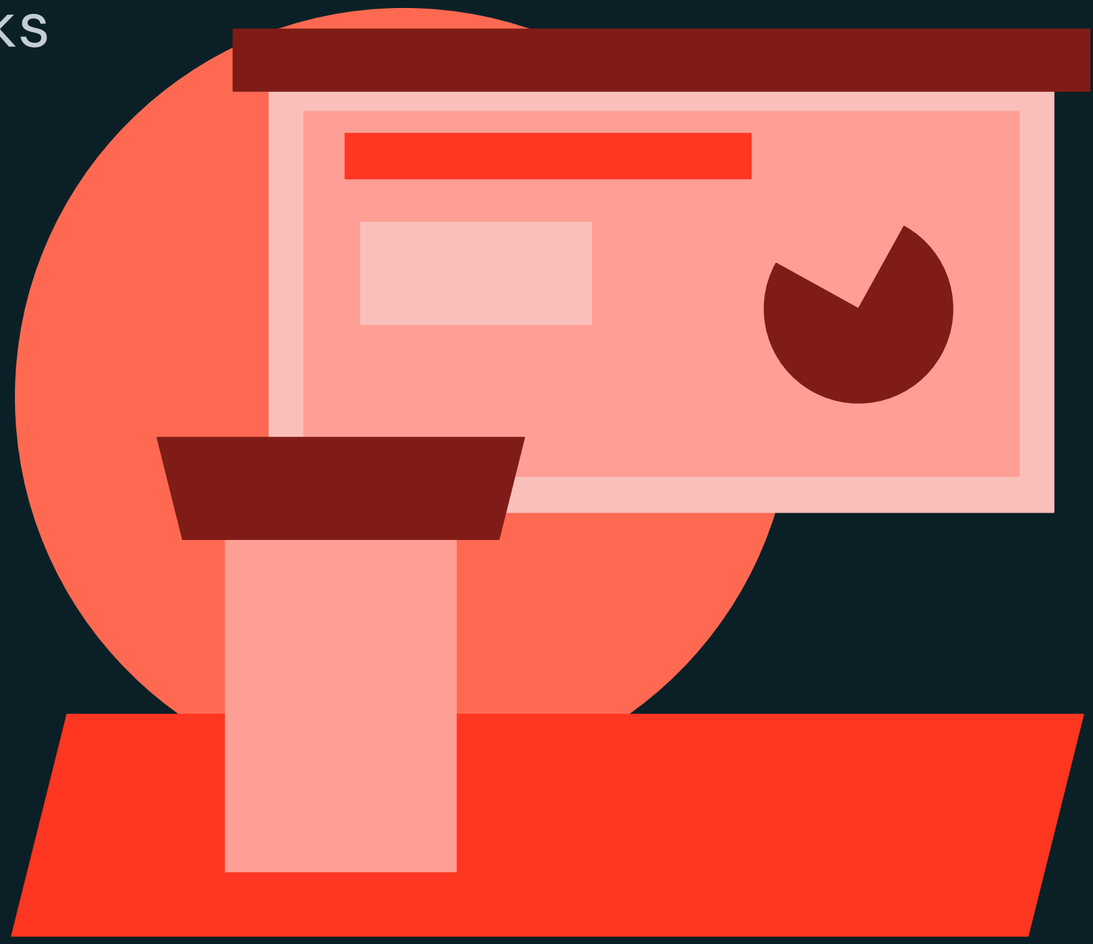
| Section Overview – Introduction to Data Engineering in Databricks |
| --- |
| • Data Engineering in Databricks |
| • What are Lakeflow Spark Declarative Pipelines? |
| • Course Setup and Creating a Pipeline |
| • Course Project Overview |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Introduction to Data Engineering in Databricks

LECTURE

# Data Engineering in Databricks



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Data Engineering in Databricks

## databricks LAKEFLOW

### UNIFIED DATA ENGINEERING FOR THE DATA INTELLIGENCE PLATFORM

**CONNECT**

Efficient ingestion connectors

**SPARK DECLARATIVE PIPELINES**
(Previously DLT)

Accelerated ETL development

**JOBS**
(Previously Workflows)

Reliable orchestration for analytics and AI

**INDUSTRY LEADING DATA PROCESSING ENGINE**
(Apache Spark + Structured Streaming)

| | |
|---|---|
| UNIFIED GOVERNANCE | Unity Catalog |
| OPTIMIZED STORAGE | DELTA LAKE Parquet ICEBERG |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Data Engineering in Databricks

**Simplifying** batch and streaming ETL with automated reliability, optimizations and built-in data quality with **Lakeflow Spark Declarative Pipelines (SDP)**!

## databricks LAKEFLOW

### UNIFIED DATA ENGINEERING FOR THE DATA INTELLIGENCE PLATFORM

**CONNECT**

Efficient ingestion connectors

**SPARK DECLARATIVE PIPELINES**
(Previously DLT)

Accelerated ETL development

**JOBS**
(Previously Workflows)

Reliable orchestration for analytics and AI

**INDUSTRY LEADING DATA PROCESSING ENGINE**
(Apache Spark + Structured Streaming)

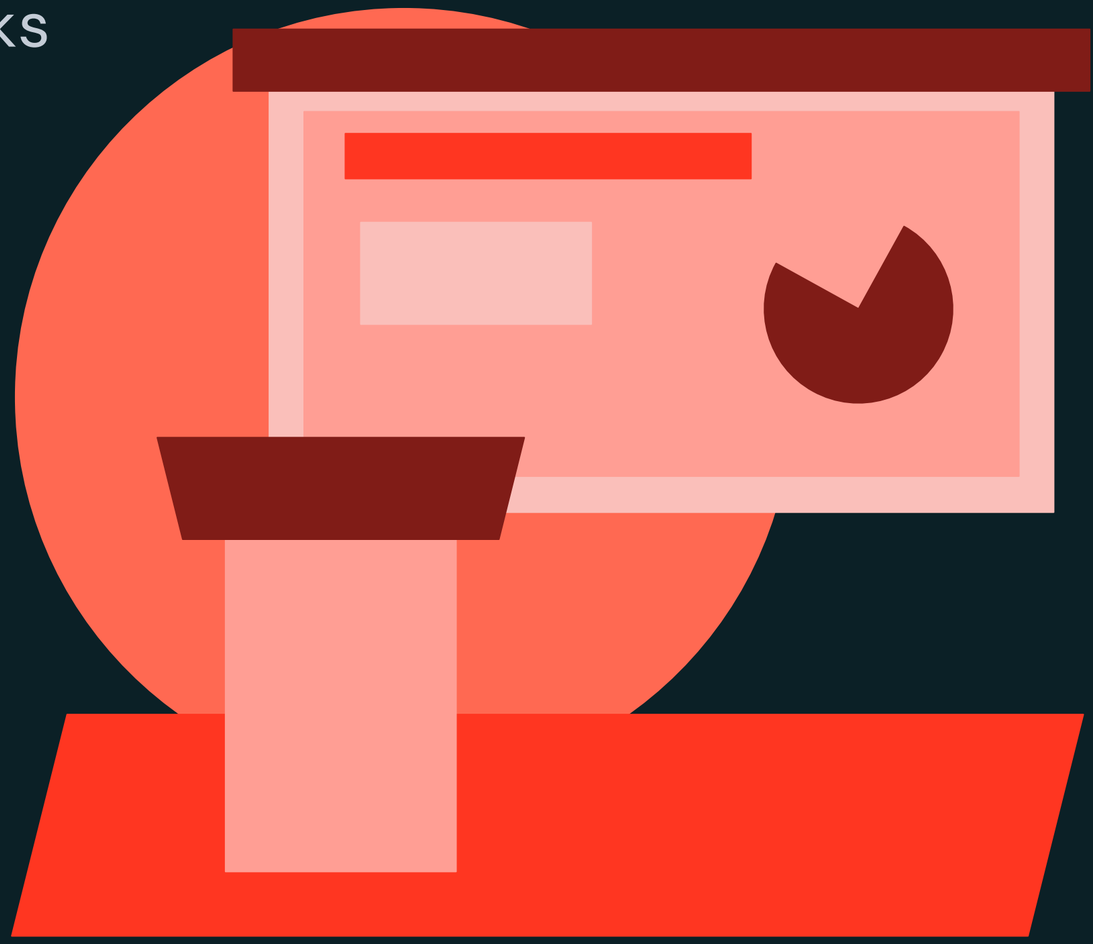
| | |
|---|---|
| UNIFIED GOVERNANCE | Unity Catalog |
| OPTIMIZED STORAGE | DELTA LAKE Parquet ICEBERG |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Introduction to Data Engineering in Databricks

LECTURE

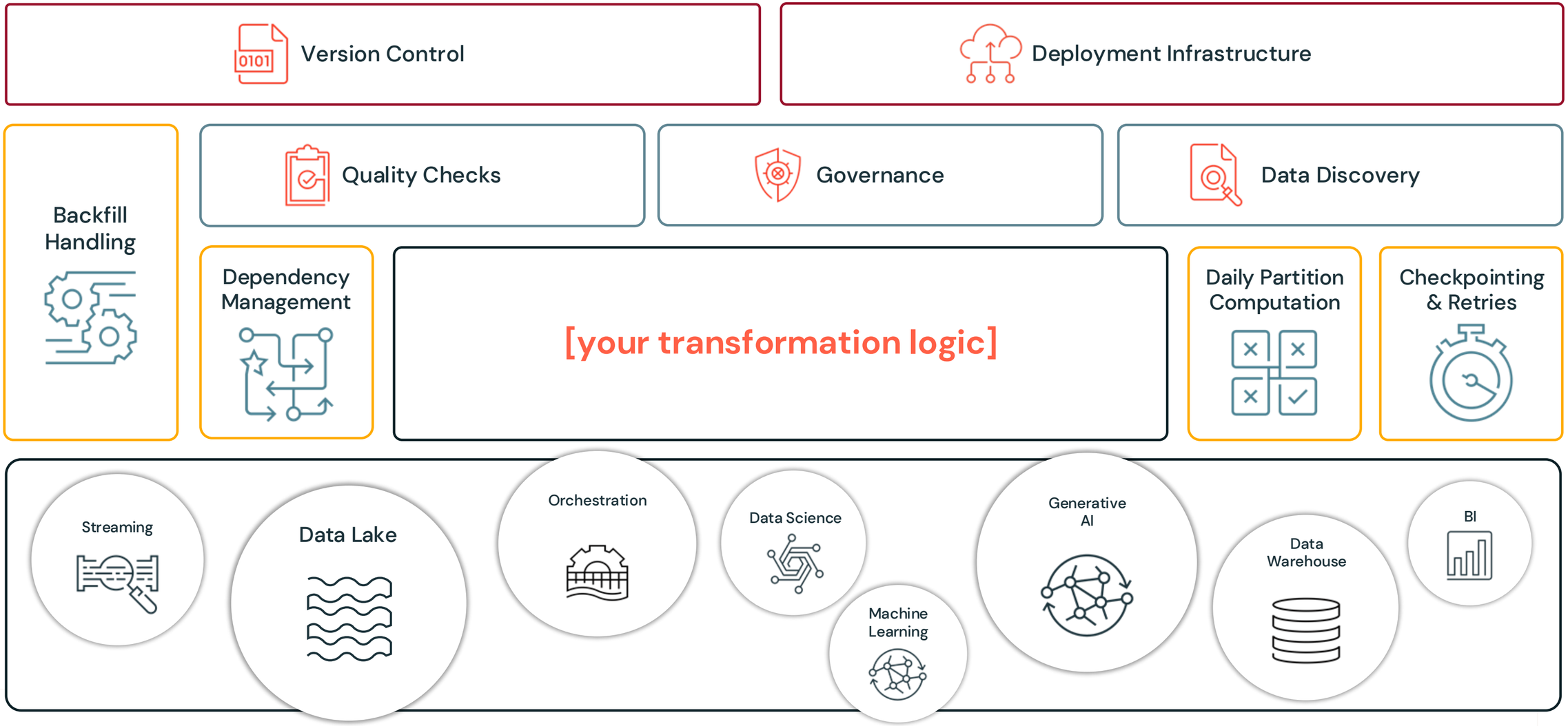
# What are Lakeflow Spark Declarative Pipelines?



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

## It's difficult to build and operate reliable data pipelines




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

## It's difficult to build and operate reliable data pipelines

**Labor-Intensive Development**

**Operational Complexity**

**Siloed Batch and Streaming**

**Slow & Error-Prone Pipelines:**
Delays data products and insights, slowing business impact.

**High Operational Costs:**
Inefficiencies drive downtime, engineering toil, and wasted resources.

**Limited Flexibility:**
Adapting to evolving latency, cost, and SLA needs is cumbersome and expensive.

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

## Spark Declarative Pipelines: Reliable data pipelines made easy

Version Control

Deployment Infrastructure

Backfill Handling

Quality Checks

Governance

Data Discovery

Dependency Management

**With Spark Declarative Pipelines simply focus on**

**[your transformation logic]**

Daily Partition Computation

Checkpointing & Retries

Streaming

Data Lake

Orchestration

Data Science

Machine Learning

Generative AI

Data Warehouse

BI

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

## Introducing Declarative Pipelines: Reliable data pipelines made easy

- **Simplified Pipeline Authoring**

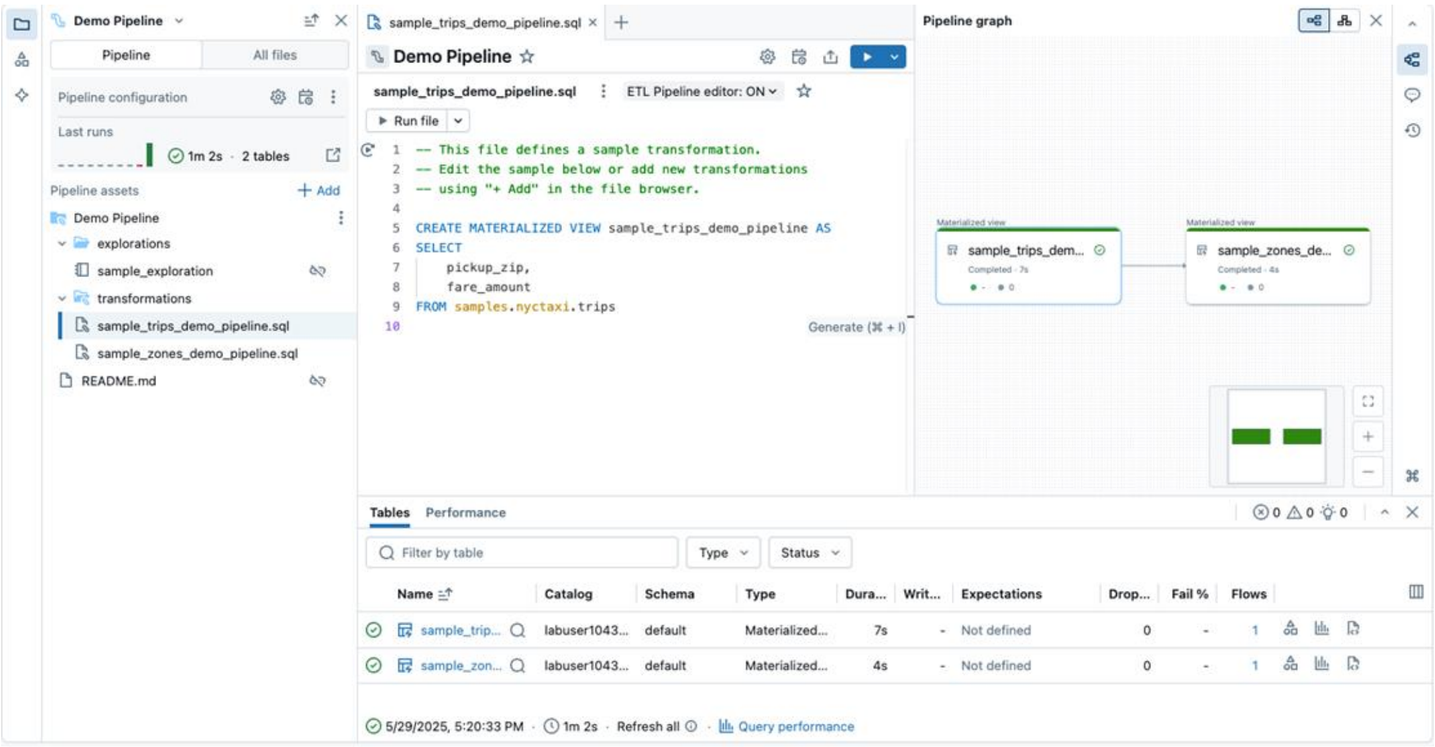
  Easily declare data ingestion and transformations with **SQL or Python,** and let Spark Declarative Pipelines handle the rest!

- **Intelligent Optimization at Scale**

  **Automated scaling and recovery** improve reliability and reduce maintenance.

- **Unified Batch and Streaming**

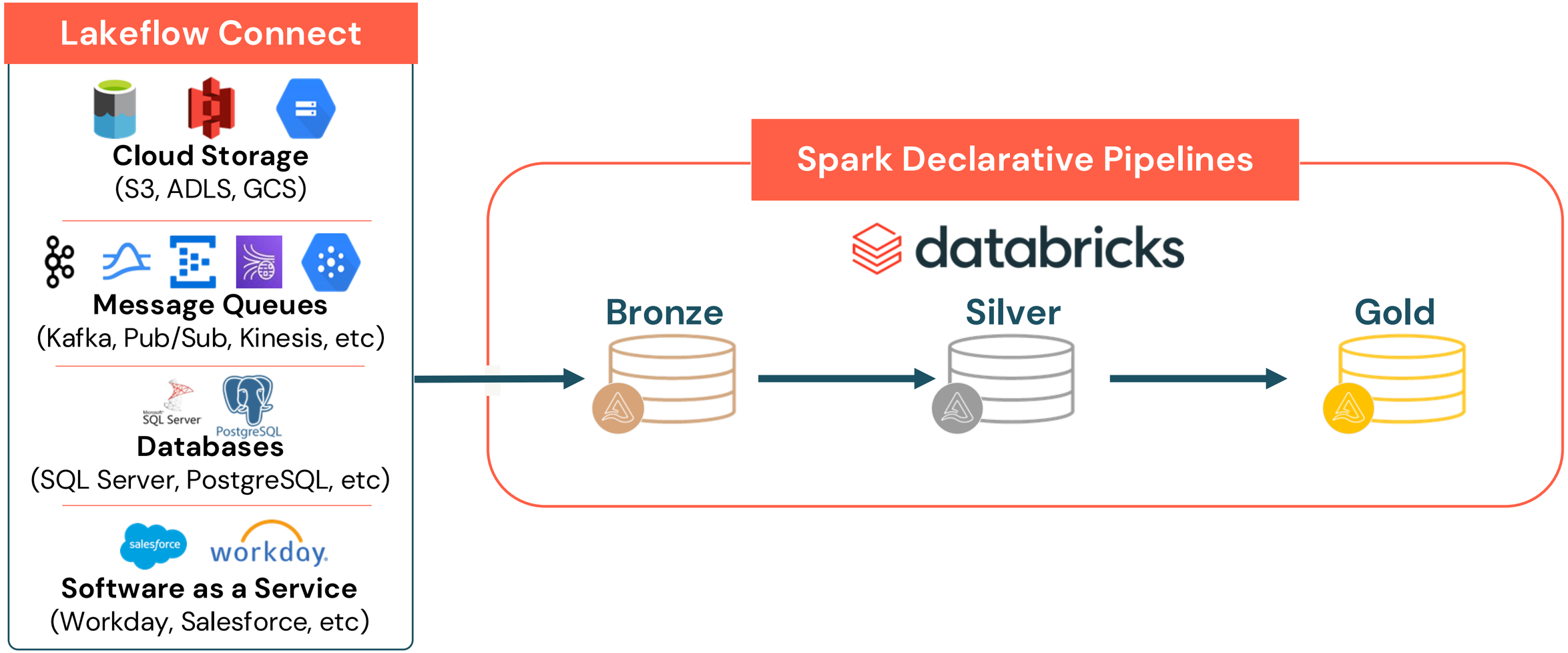
  Pipelines seamlessly adapt to **both near real-time and batch Lakeflow Jobs,** optimizing performance and cost.




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

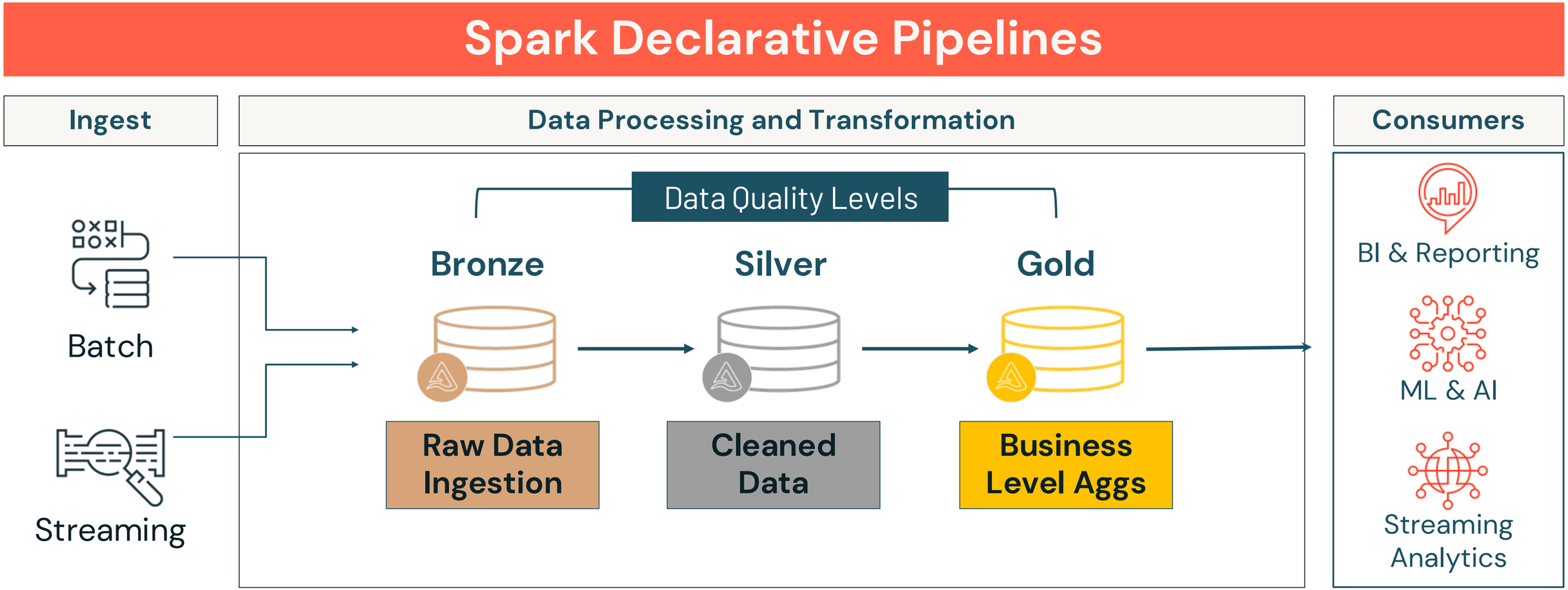
## Connecting to Data Sources




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

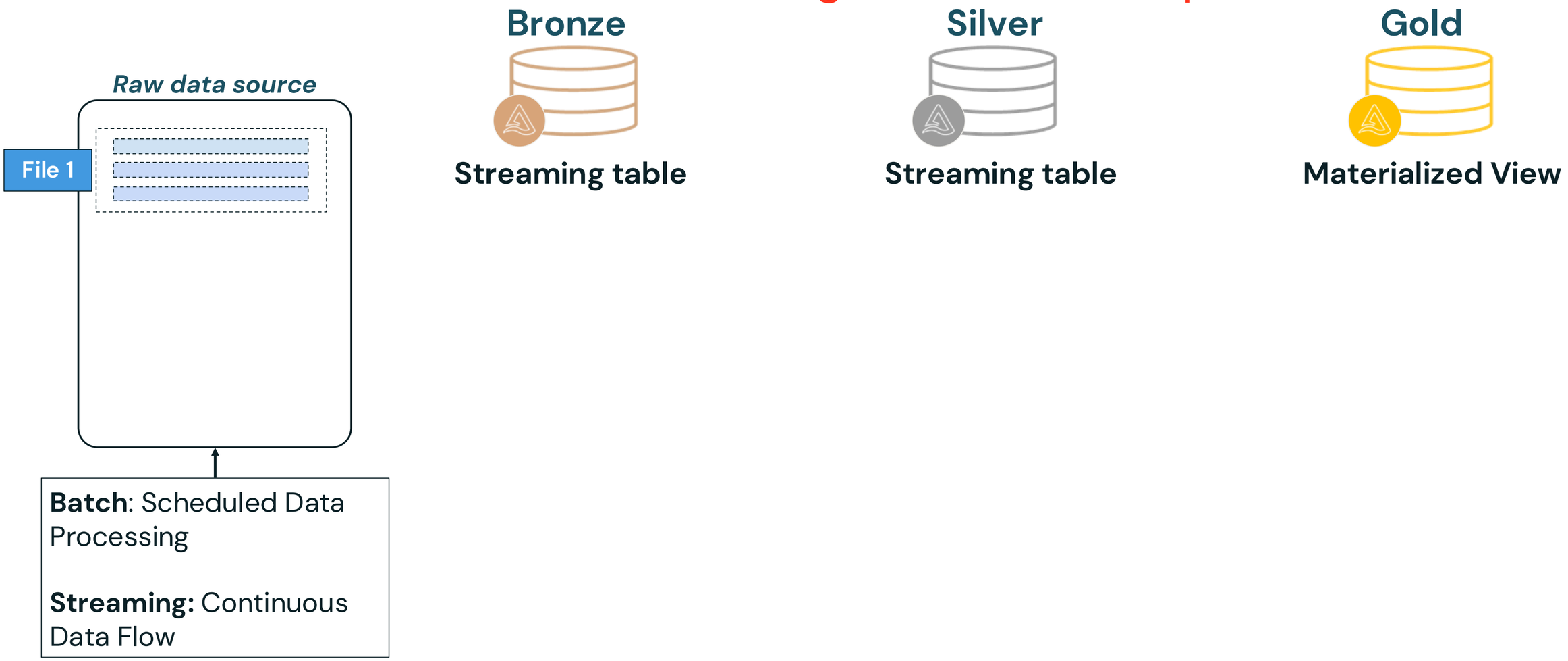
## Simplifying Batch and Streaming ETL in the Medallion Architecture




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

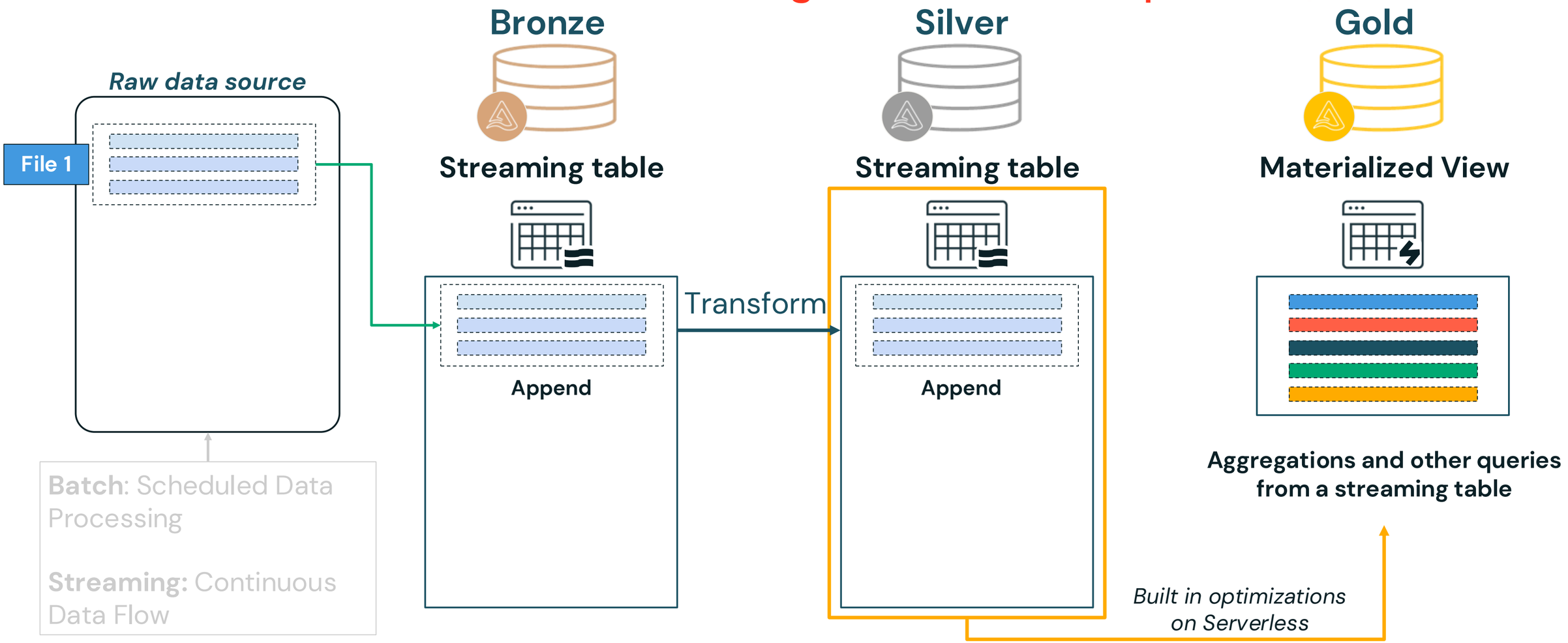
## Overview of Incremental Processing in Declarative Pipelines




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# What are Spark Declarative Pipelines?

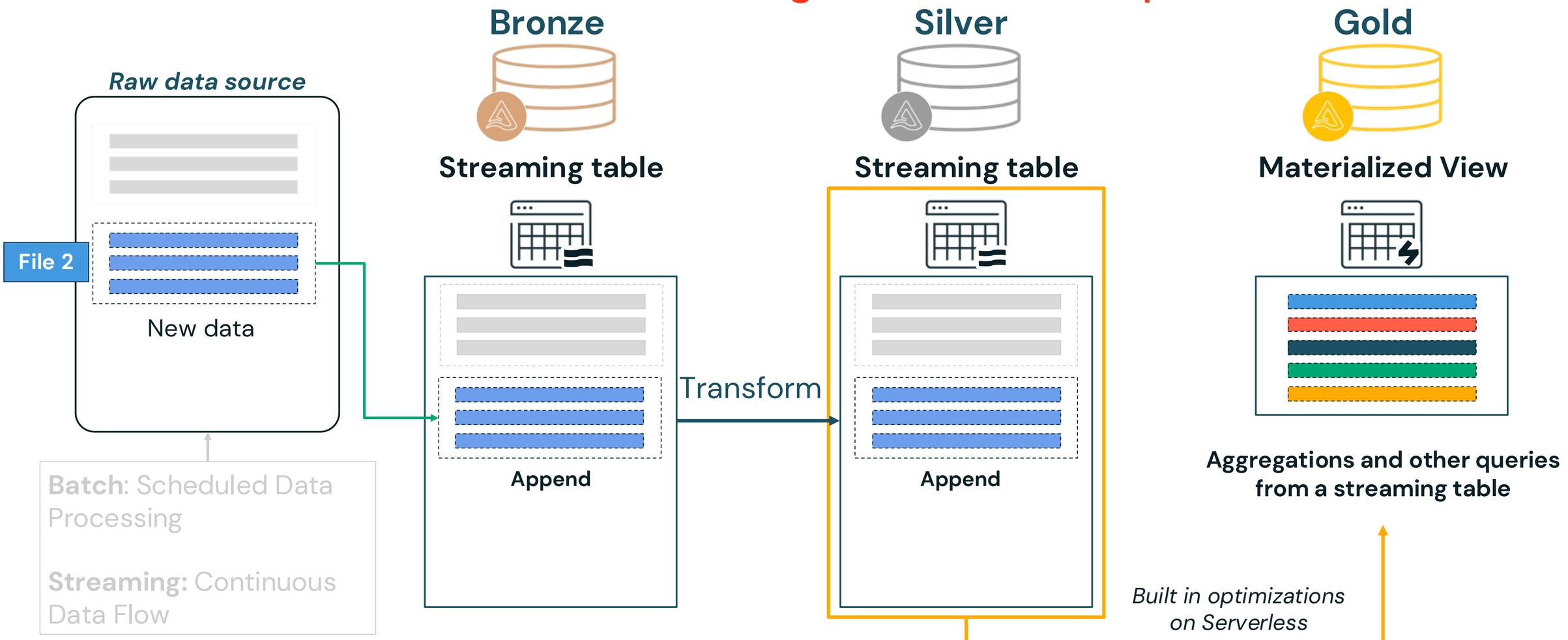
## Overview of Incremental Processing in Declarative Pipelines (Run 1)




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

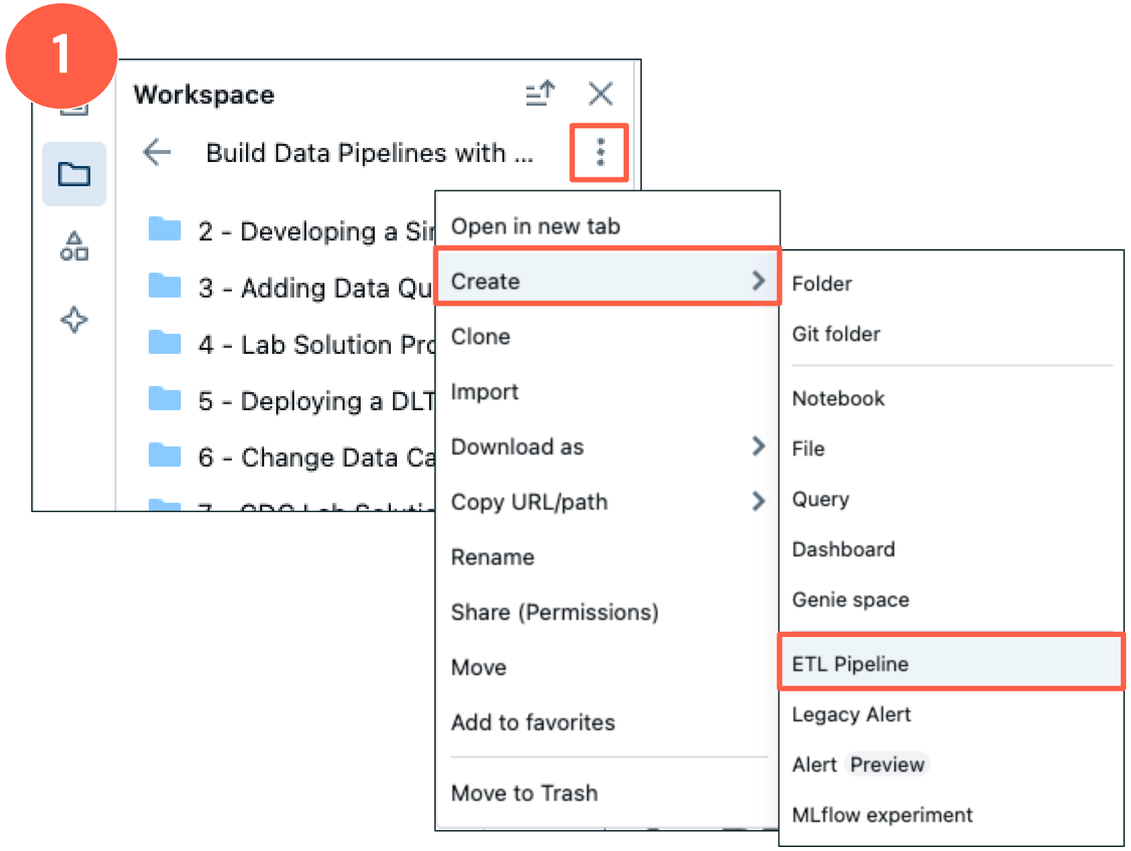
# What are Spark Declarative Pipelines?

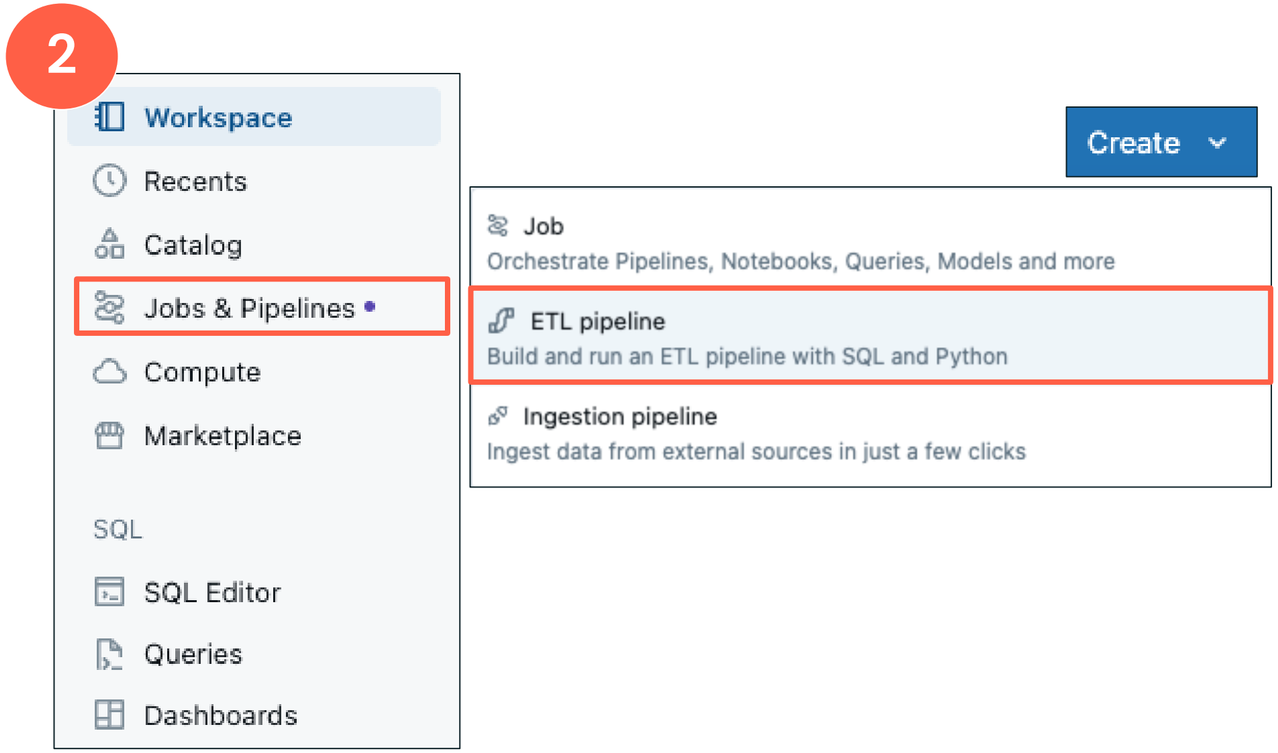
## Overview of Incremental Processing in Declarative Pipelines (Run 2)




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Creating a Spark Declarative Pipeline



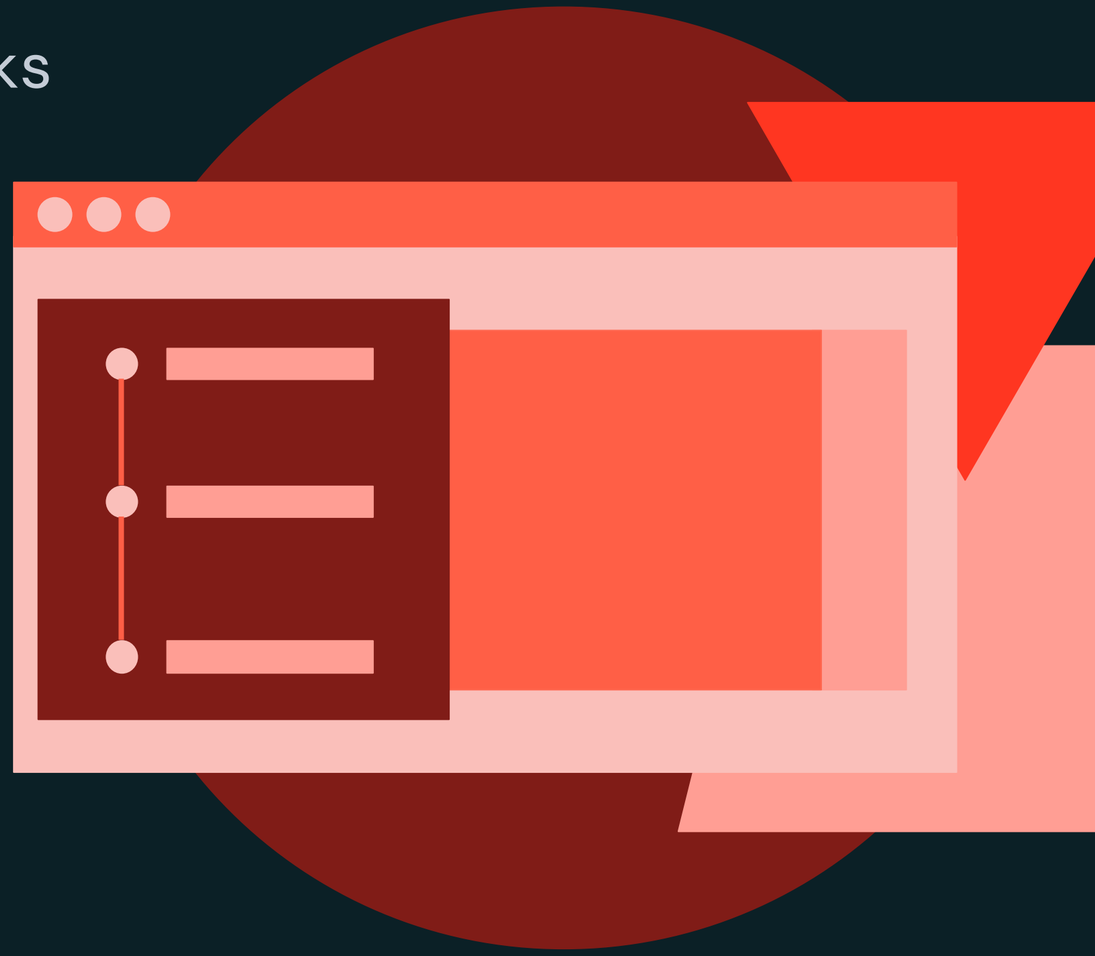




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Introduction to Data Engineering in Databricks

DEMONSTRATION

# REQUIRED – Course Setup and Creating a Pipeline



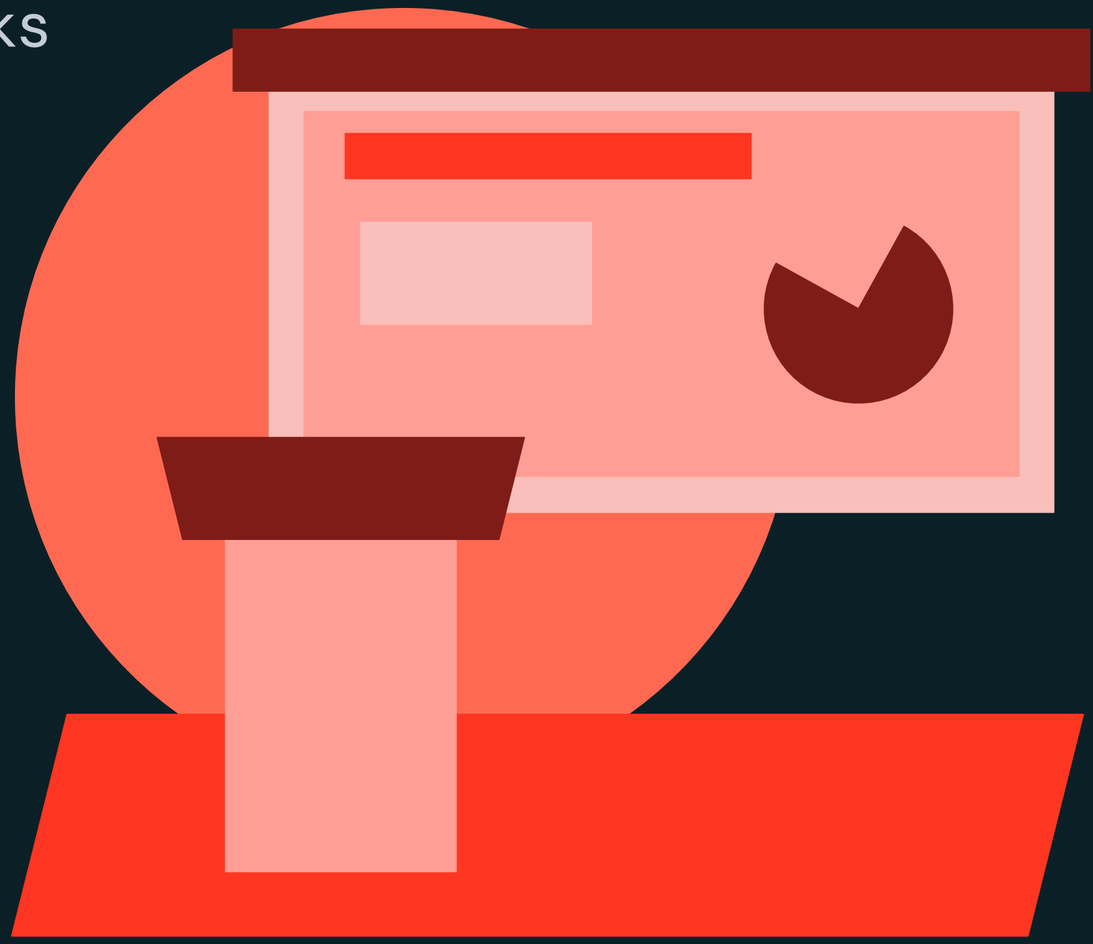
Notebook: 1 – REQUIRED – Course Setup and Creating a Pipeline

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Introduction to Data Engineering in Databricks

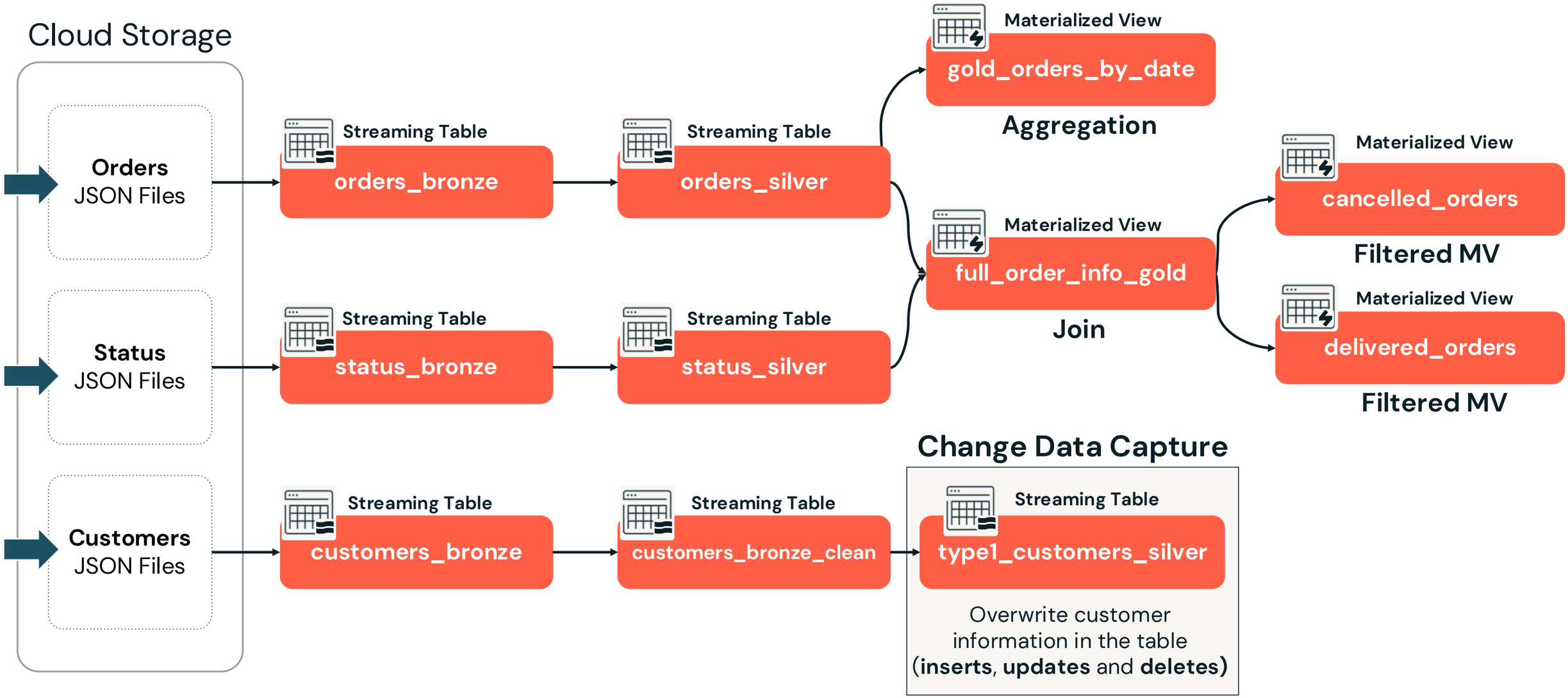
LECTURE

# Course Project Overview



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Course Project Overview




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Lakeflow Spark Declarative Pipeline Fundamentals

**Build Data Pipelines with Lakeflow Spark Declarative Pipelines**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Section Learning Objectives

- Understand the core concepts and components of Lakeflow Spark Declarative Pipelines pipelines, including how streaming tables, materialized views, and temporary views function and differ.
- Identify and configure pipeline settings such as compute, data assets, trigger modes, and advanced options.
- Develop a functional Declarative Pipeline using the new pipeline editor and SQL-based syntax for the pipeline.
- Incorporate data quality expectations into a Declarative Pipeline pipeline to validate and enforce data integrity.

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Agenda

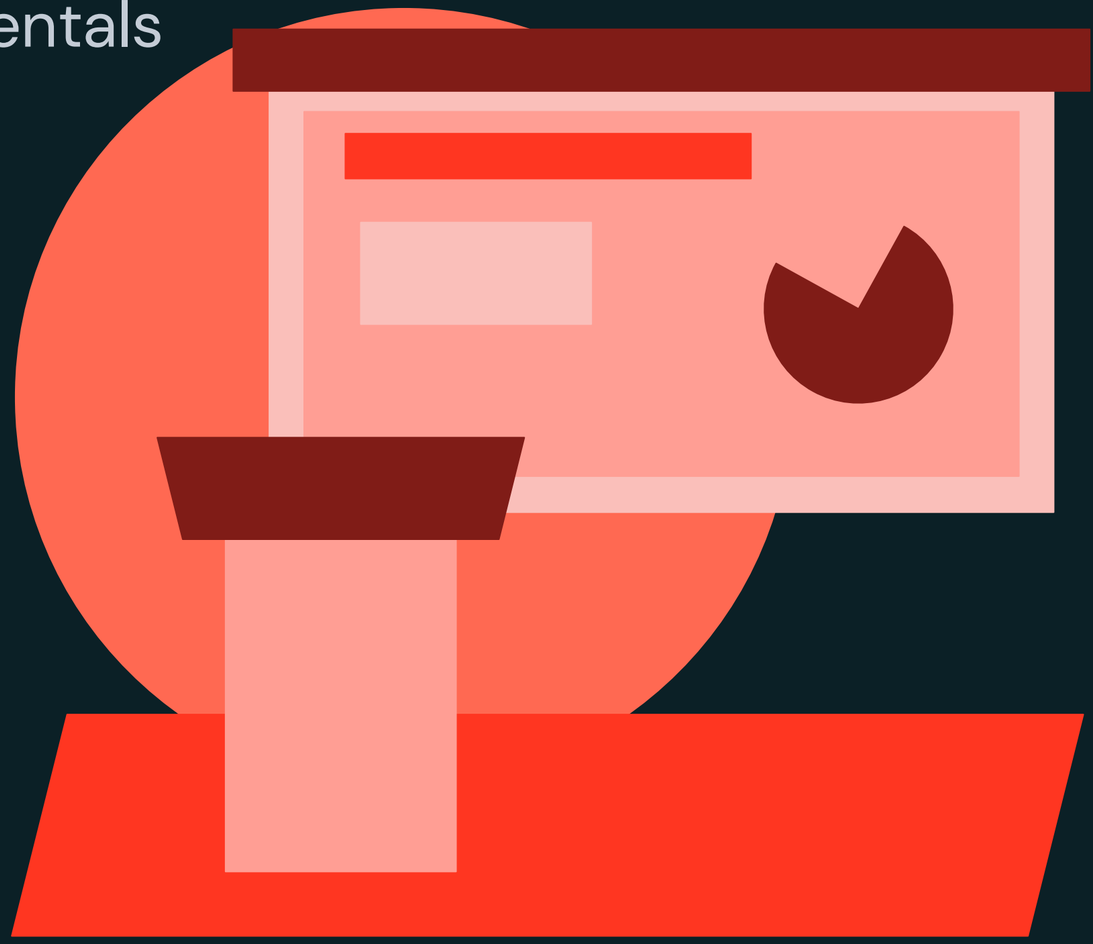
| Section Overview – Lakeflow Spark Declarative Pipeline Fundamentals |
| --- |
| • Dataset Types Overview |
| • Simplified Pipeline Development |
| • Common Pipeline Settings |
| • Developing a Simple Pipeline |
| • Ensure Data Quality with Expectations |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

# Dataset Types Overview



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## How streaming tables, materialized views, and views process data

### STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**

### MATERIALIZED VIEW (MV)

### VIEWS

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## STREAMING TABLE Overview

### STREAMING TABLE

- Support for **batch** or **streaming** for **incremental** data processing (exactly **once processing** of the files)
- Each time an streaming table is refreshed, data added to the source tables is **appended** to the **streaming table**
- SQL code that creates streaming tables:
  `CREATE OR REFRESH STREAMING TABLE`
- In SQL, you must use `FROM STREAM read_files()` to **invoke Auto Loader functionality** for incremental streaming reads and checkpoints
- File names are **guaranteed to be read only once**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## Create a STREAMING TABLE with SQL

Create a STREAMING TABLE named **orders_bronze** from **JSON files**

```sql
CREATE OR REFRESH STREAMING TABLE 1_bronze_db.orders_bronze AS
SELECT
 *,
 current_timestamp() AS processing_time,
 _metadata.file_name AS source_file
FROM STREAM read_files(
   "/Volumes/dbacademy/ops/labuser/orders",
   format => 'JSON');
```

*Note: The query is using the default catalog*

Creates a streaming table named **orders_bronze** in the **1_bronze_db** schema

Use the SELECT clause to specify the columns to read into the streaming table from your data source

- The **STREAM** keyword indicates to use streaming read for the source files (JSON)
- The **read_files()** function reads files under a provided location and returns the data in tabular form

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## Create a STREAMING TABLE with SQL

Create a STREAMING TABLE named **orders_silver** that reads from the **STREAMING orders_bronze** table

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver AS
SELECT
    order_id,
  timestamp(order_timestamp) AS order_timestamp,
  customer_id,
  notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

Specify your SQL transformations for the streaming data from the bronze table

Use **STREAM** keyword with the name of the STREAMING TABLE you are reading from to transform new rows

*Note: The query is using the default catalog*

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## How streaming tables, materialized views, and views process data

### STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**

### MATERIALIZED VIEW (MV)

- Records are processed as required to return accurate results for the **current data state**
- Used for data processing tasks such as:
  - Transformations
  - Aggregations
  - Pre-computing slow queries
  - Frequently used computations

### VIEWS

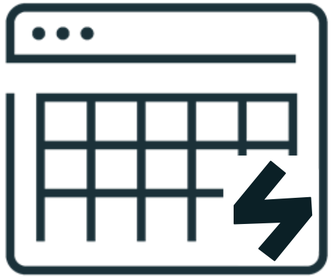
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## MATERIALIZED VIEW Overview

### MATERIALIZED VIEW

- Each time a materialized view is updated, **query results are recalculated to reflect changes in upstream datasets**
- Created and **kept up-to-date** by pipeline
- Use the `CREATE OR REFRESH MATERIALIZED VIEW` syntax
- Can be used **anywhere in your pipeline**, not just in the gold layer
- **Where applicable,** results are **incrementally refreshed for materialized views**, avoiding the need to completely rebuild the materialized view when new data arrives **(Serverless Only)**
- Incremental refresh for materialized views is a cost-based optimizer to power fast and efficient transformations for materialized views on Serverless compute



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## Create a MATERIALIZED VIEW with SQL

Create a MATERIALIZED VIEW named **gold_orders_by_date** that summarizes the **orders_silver** table

```
CREATE OR REFRESH MATERIALIZED VIEW 3_gold_db.gold_orders_by_date AS
SELECT
 date(order_timestamp) AS order_date,
 count(*) AS total_daily_orders
FROM 2_silver_db.orders_silver
GROUP BY date(order_timestamp);
```

Create the materialized view **gold_orders_by_date** in the **3_gold_db** schema (database)

Specify the the source table **without the STREAM** keyword to utilize the source STREAMING table
***NOTE: Where possible, queries will be incrementally refreshed***

*Note: The query is using the default catalog*

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## How streaming tables, materialized views, and views process data

### STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**

### MATERIALIZED VIEW (MV)

- Records are processed as required to return accurate results for the **current data state**
- Used for data processing tasks such:
  - Transformations
  - Aggregations
  - pre-computing slow queries
  - frequently used computations.

### VIEWS

Constructs a virtual table with no physical data based on the query in your Declarative Pipelines

- TEMPORARY VIEW
- VIEW

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## TEMPORARY VIEW Overview

### TEMPORARY VIEW

- Temporary views are only **persisted across the lifetime** of the pipeline and **private to the defining pipeline**
- They are **not registered** as an object to Unity Catalog
- Use the `CREATE TEMPORARY VIEW` statement
- TEMPORARY VIEWS are useful as **intermediate queries** that **are not exposed** to end users

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

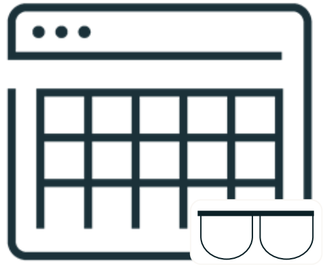
## VIEW Overview

### VIEW

- Constructs a virtual table with no physical data based on the result-set of a SQL query in your pipelines
- They are **registered** as an object to Unity Catalog
- Use the `CREATE VIEW` statement

**LIMITATIONS**

- The pipeline must be a Unity Catalog pipeline
- Views cannot have streaming queries, or be used as a streaming source for a pipeline



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

Existing users of DLT will notice that we've evolved the names

| | *Deprecated* |
|---|---|
| `CREATE OR REFRESH STREAMING TABLE` | `CREATE OR REFRESH STREAMING LIVE TABLE` |
| `CREATE OR REFRESH MATERIALIZED VIEW` | `CREATE OR REFRESH LIVE TABLE` |
| `CREATE VIEW / CREATE TEMPORARY VIEW` | `CREATE LIVE VIEW / CREATE TEMPORARY LIVE VIEW` |

The **semantics remain the same**, and **we'll support the old syntax** for compatibility. Our goal is to simplify the syntax and match other systems

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Dataset Types Overview

## Pipeline dependencies are parsed automatically, code order is not required

### Raw JSON -> Bronze

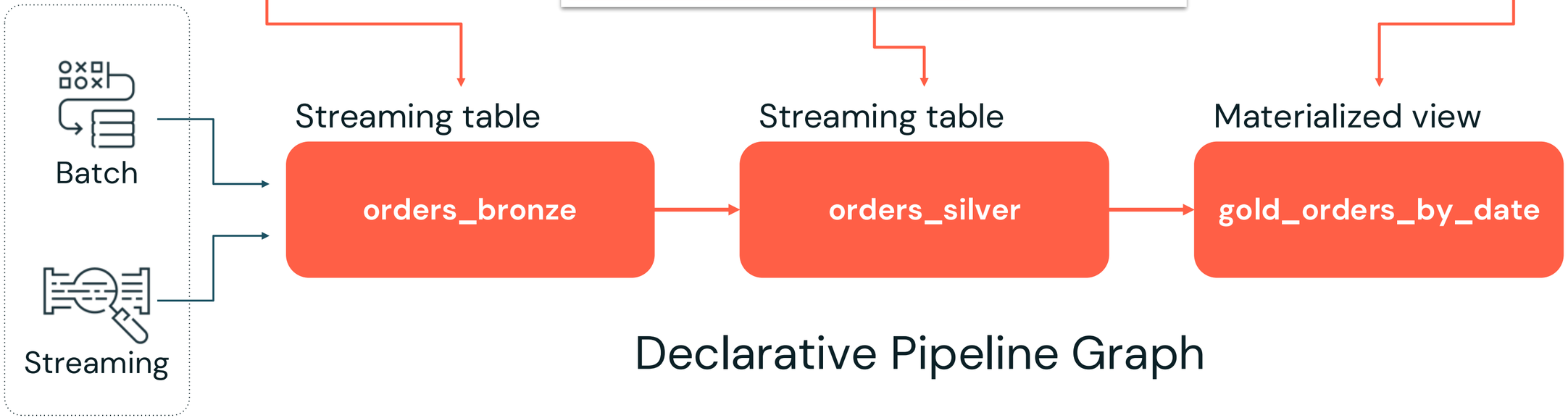
```
CREATE OR REFRESH STREAMING TABLE
1_bronze_db.orders_bronze AS
SELECT
 *,
 current_timestamp() AS processing_time,
 _metadata.file_name AS source_file
FROM STREAM read_files(
   "/Volumes/dbacademy/ops/labuser/orders",
   format => 'JSON');
```

### Bronze -> Silver

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver
AS
SELECT
  order_id,
  timestamp(order_timestamp) AS order_timestamp,
  customer_id,
  notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

### Silver -> Materialized View

```
CREATE OR REFRESH MATERIALIZED VIEW 3_gold_db.gold_orders_by_date
AS
SELECT
 date(order_timestamp) AS order_date,
 count(*) AS total_daily_orders
FROM 2_silver_db.orders_silver
GROUP BY date(order_timestamp);
```



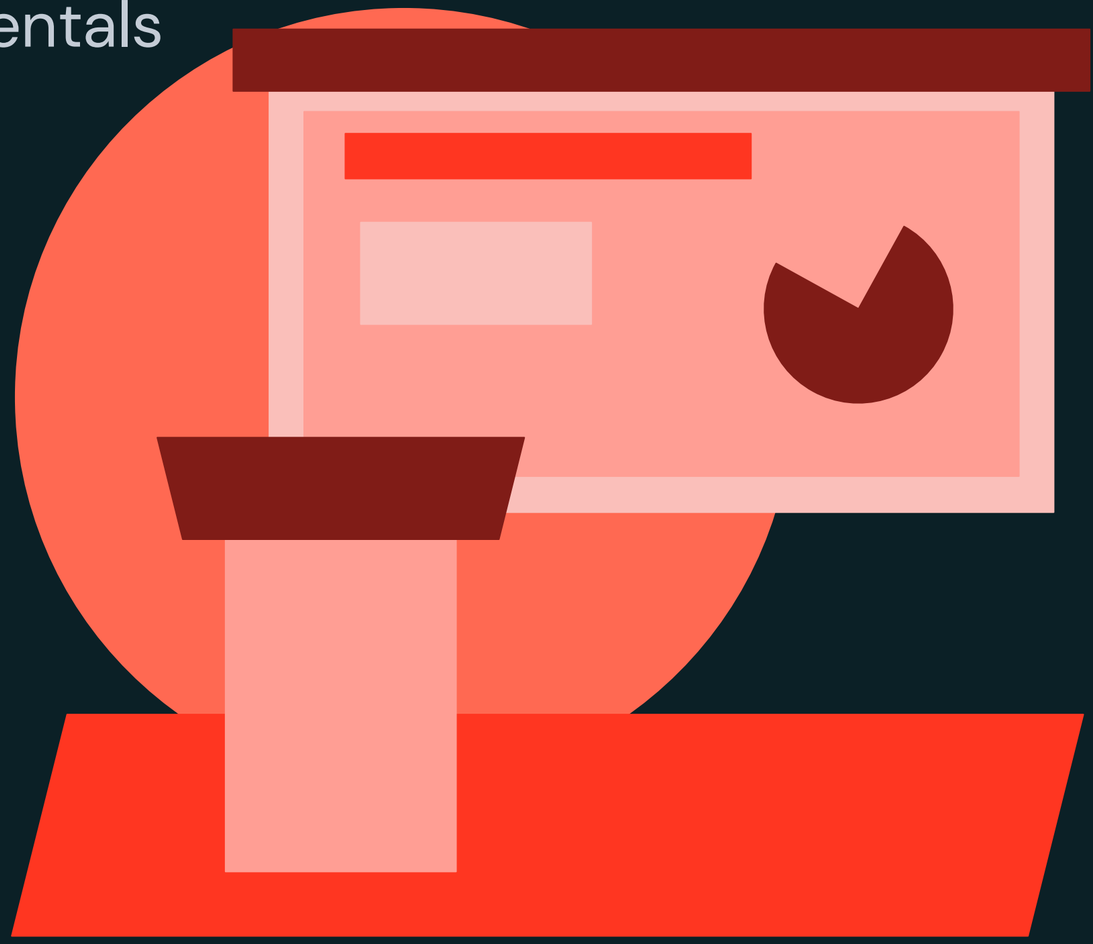

Declarative Pipeline Graph

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

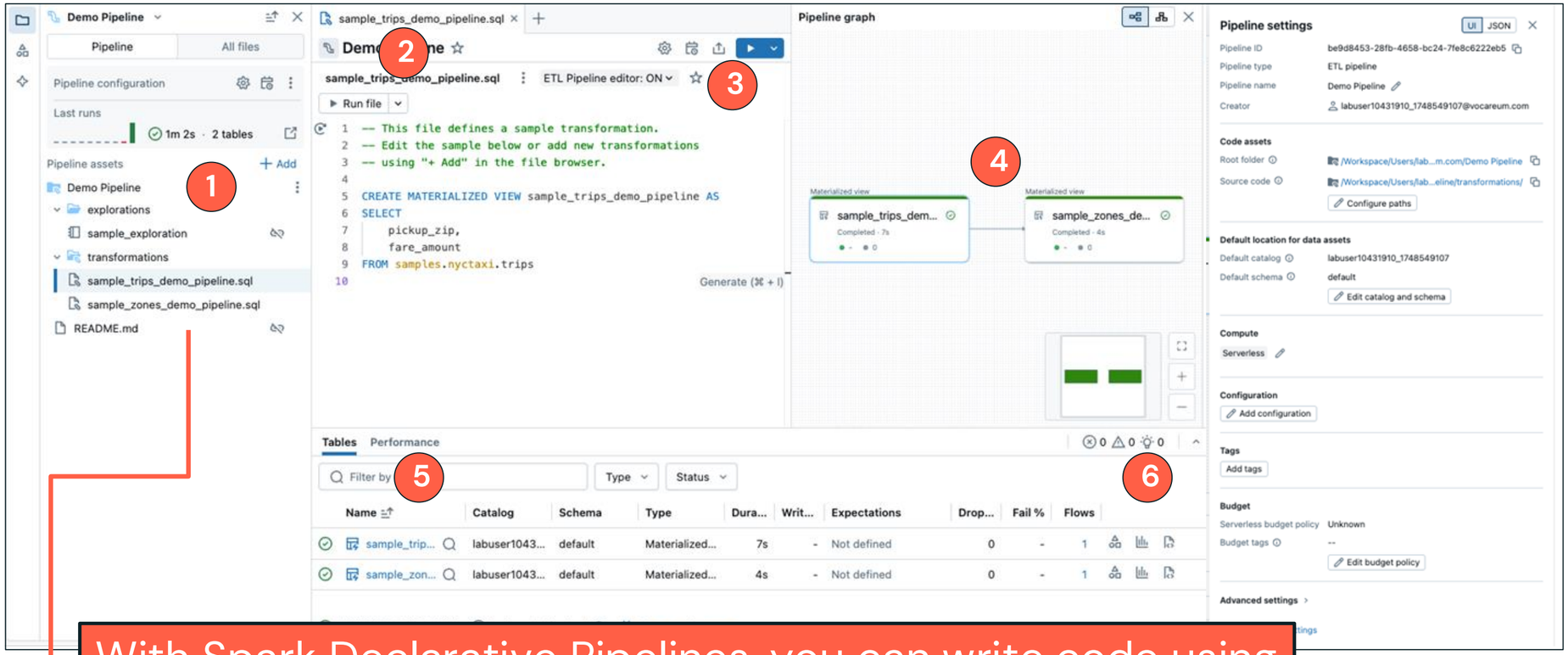
# Simplified Pipeline Development



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Simplified Pipeline Development

## Multi-file editor in Lakeflow Spark Declarative Pipelines




With Spark Declarative Pipelines, you can write code using **.sql or .py files (recommended)**, or use notebooks.

## What's New

1. Pipeline asset browser
2. Multi-file code editor with features for step-by-step development
3. Pipeline-specific toolbars
4. Interactive DAG
5. Data previews
6. Execution insights panels
   - Easier debugging
   - Faster validation with dry run

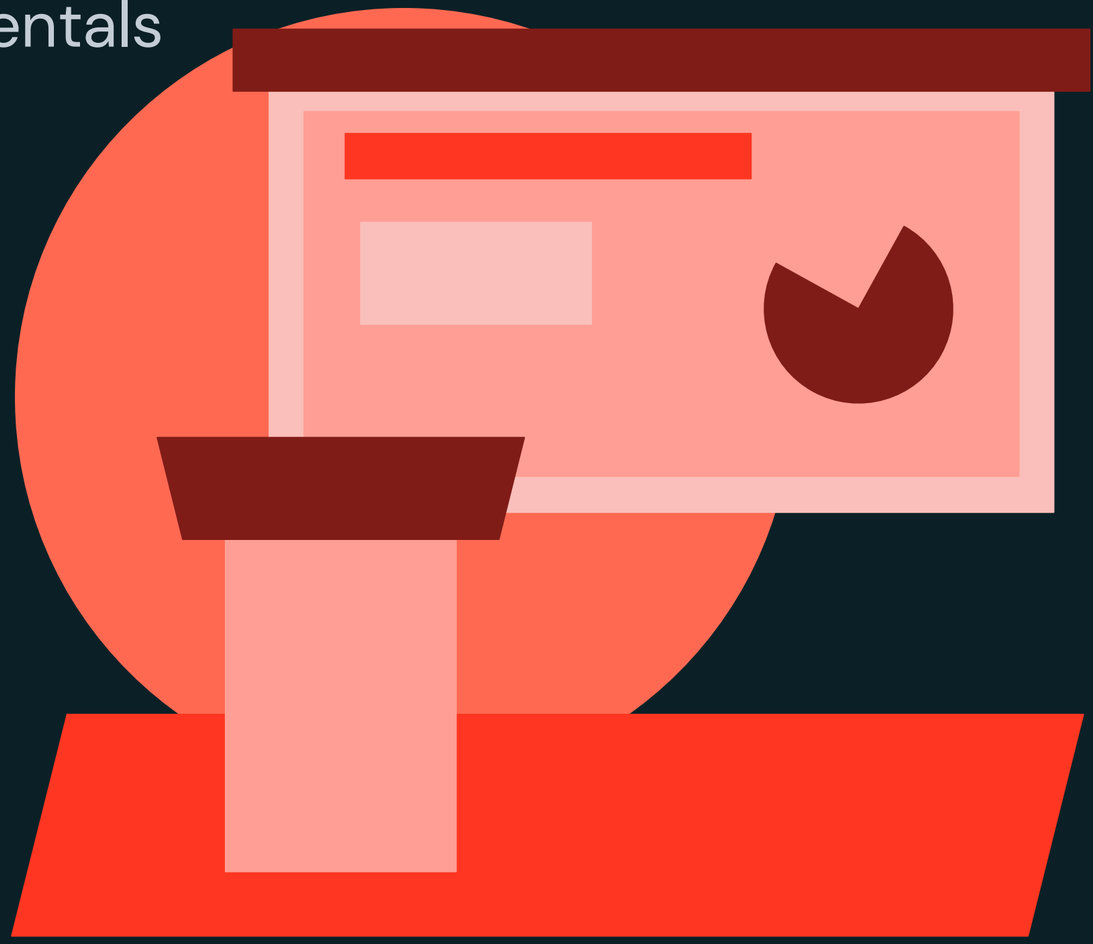
As of Data & AI Summit 2025

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

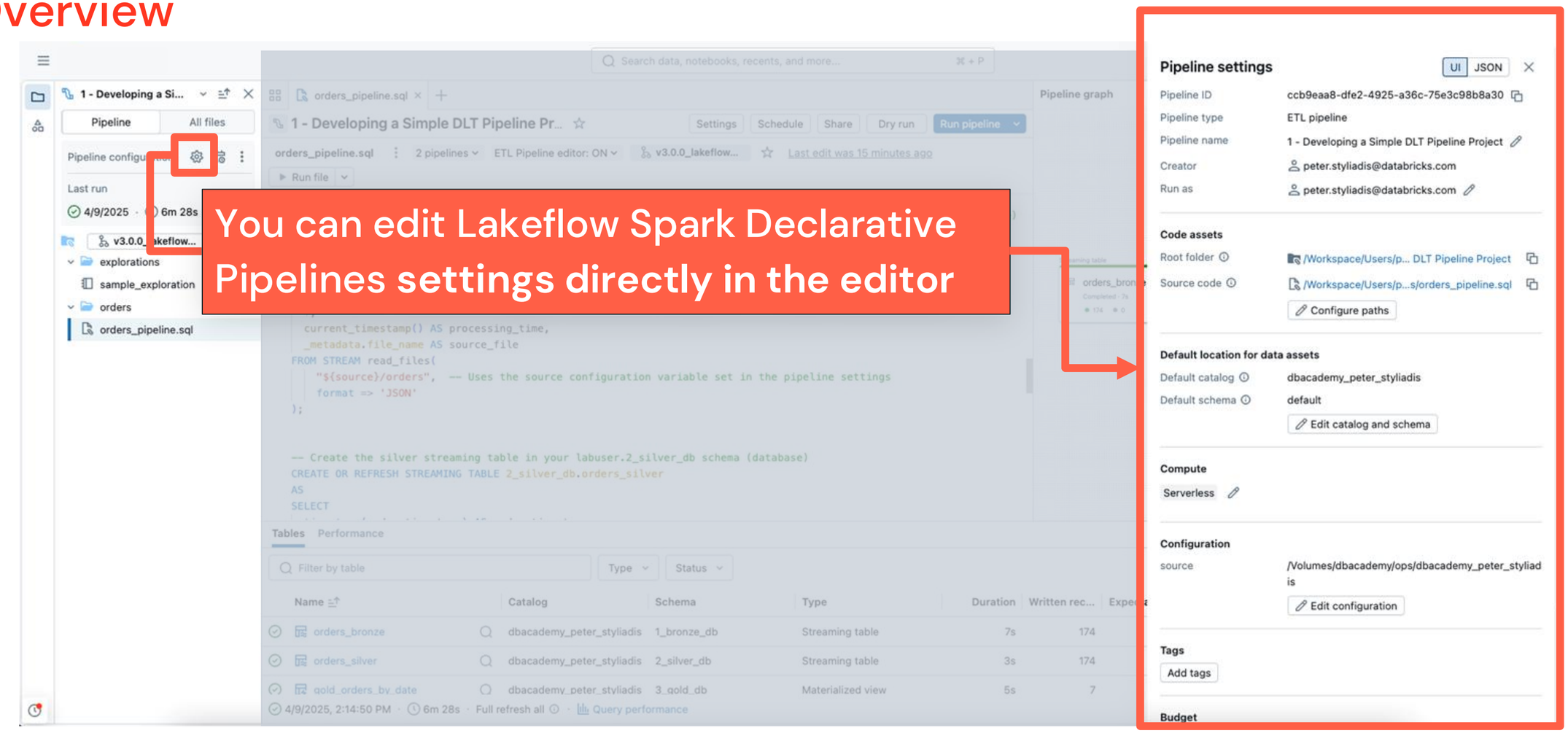
# Common Pipeline Settings



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

## Overview




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)

## SERVERLESS

- Databricks **recommends** developing new pipelines using serverless
- **Optimizes** cost while **maintaining** performance
- **Enables** you to focus on code rather than managing infrastructure
- Optional **Serverless Performance optimized** setting for time-sensitive Lakeflow Jobs

- Serverless compute includes **Incremental refresh for materialized views**
- A cost-based optimizer to power fast and efficient transformations for materialized views on Serverless compute

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)

## SERVERLESS

- Databricks **recommends** developing new pipelines using serverless
- **Optimizes** cost while **maintaining** performance
- **Enables** you to focus on code rather than managing infrastructure
- Optional **Serverless Performance optimized** setting for time-sensitive Lakeflow Jobs

## CLASSIC (fixed size)

- Users need **permission** to create compute for Declarative Pipelines
- Workspace admins **can configure cluster policies** to provide users with access to compute resources for Declarative Pipelines

## AUTOSCALING

- Enhanced autoscaling **is enabled by default** for all new pipelines.
- **Optimizes cluster utilization** by automatically allocating cluster resources based on workload volume

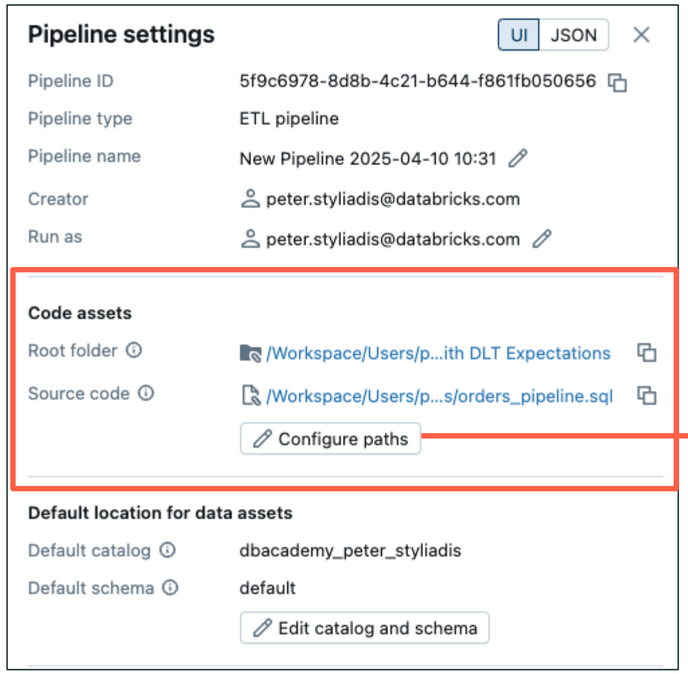
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

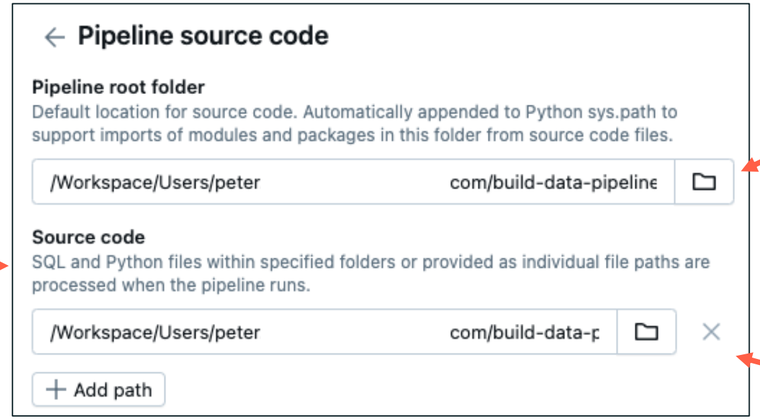
# Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)




The **Pipeline root folder** automatically includes all relevant files within the folder for the pipeline project (can be a Git folder)

The **source code** section references include subfolders (or files) for the pipeline **(.py, .sql, notebooks)**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

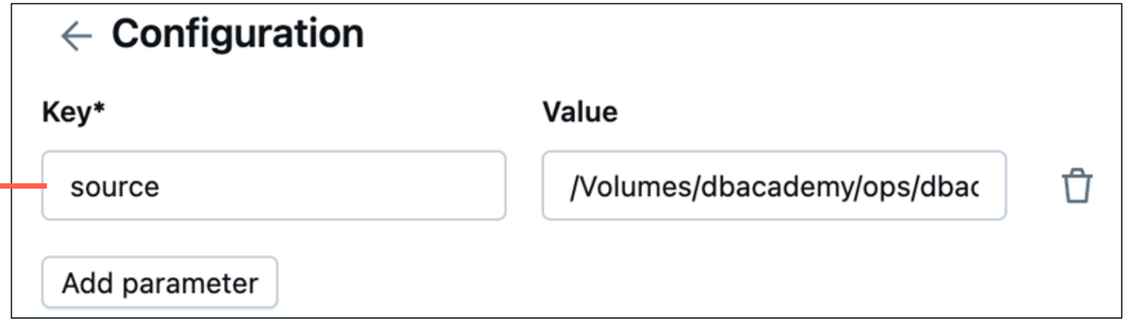
Compute

Code Assets

Configuration
(parameters)

A pipeline's configuration is **a map of key value pairs** that can be used to parameterize your code:

- Improve code readability/maintainability
- Reuse common parameters in multiple pipelines files




```
CREATE STREAMING TABLE bronze
SELECT *
FROM STREAM read_files(
  "${source}/orders",
  format => 'JSON');
```

In SQL reference the key's value using the **${key-name}** syntax

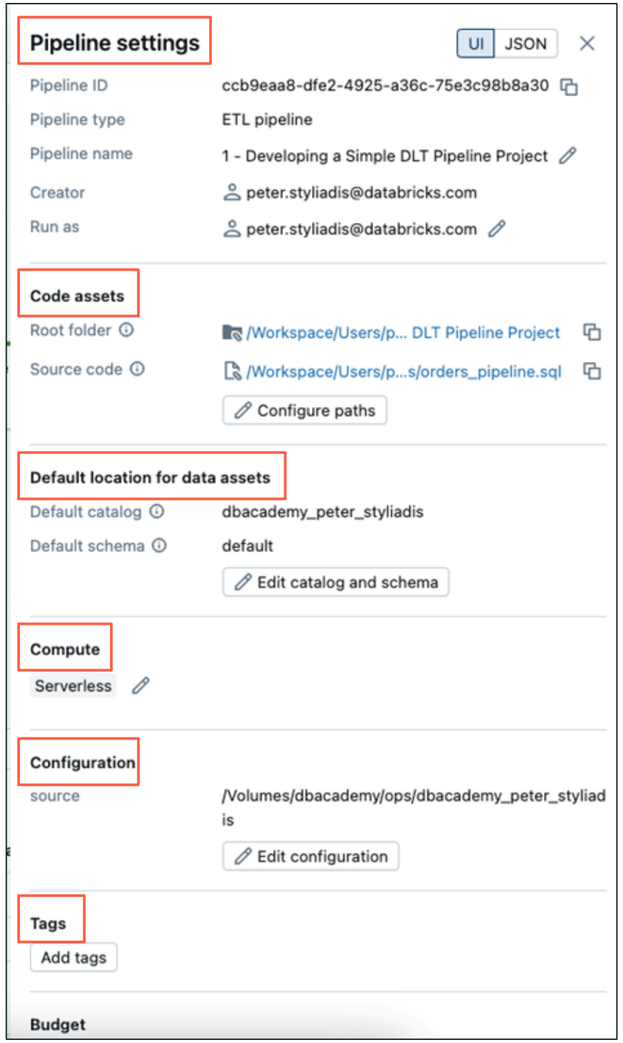
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Common Pipeline Settings

## Common Settings

There are **various other settings** we will explore throughout the course, including:

- Pipeline settings
- Code assets
- Default location for data assets
- Compute
- Configuration
- Environment
- Tags
- Budget
- Advanced settings

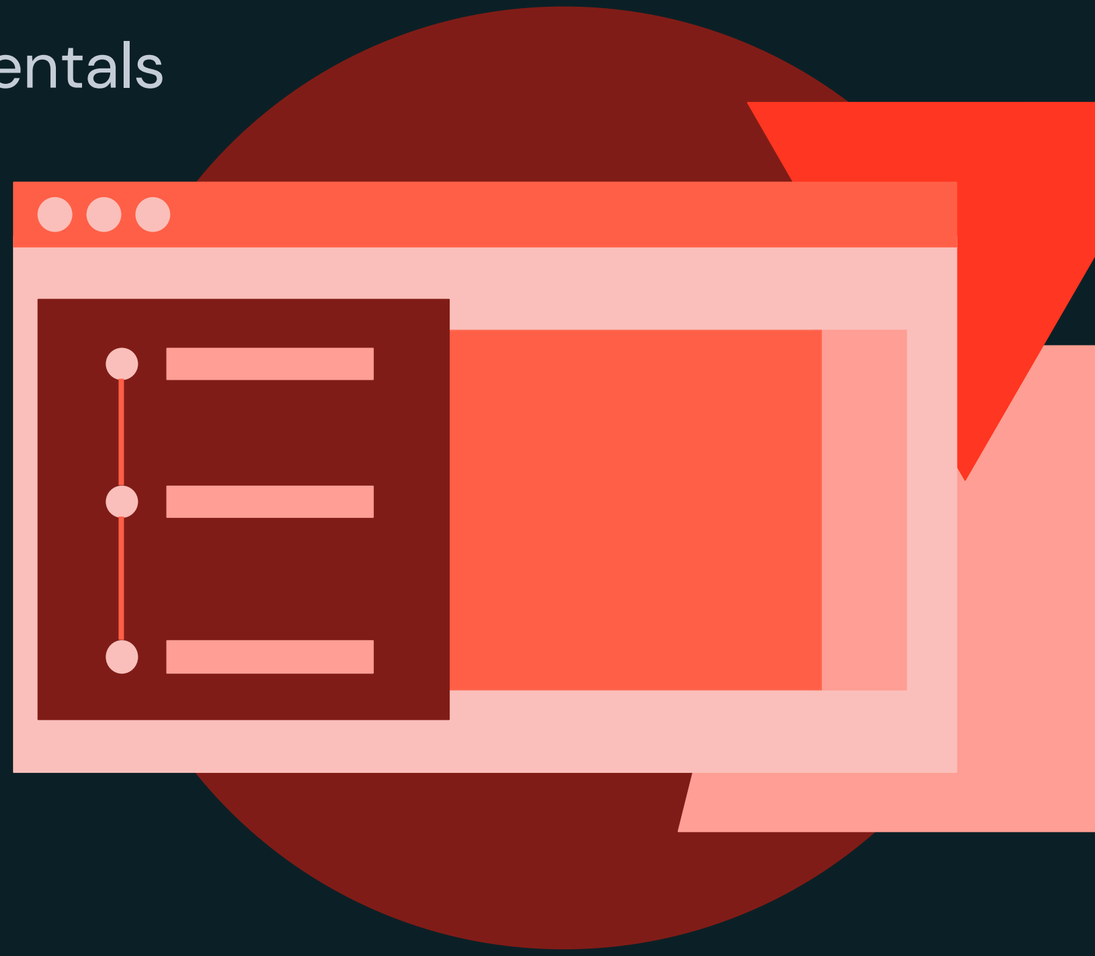



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

DEMONSTRATION

# Developing a Simple Pipeline



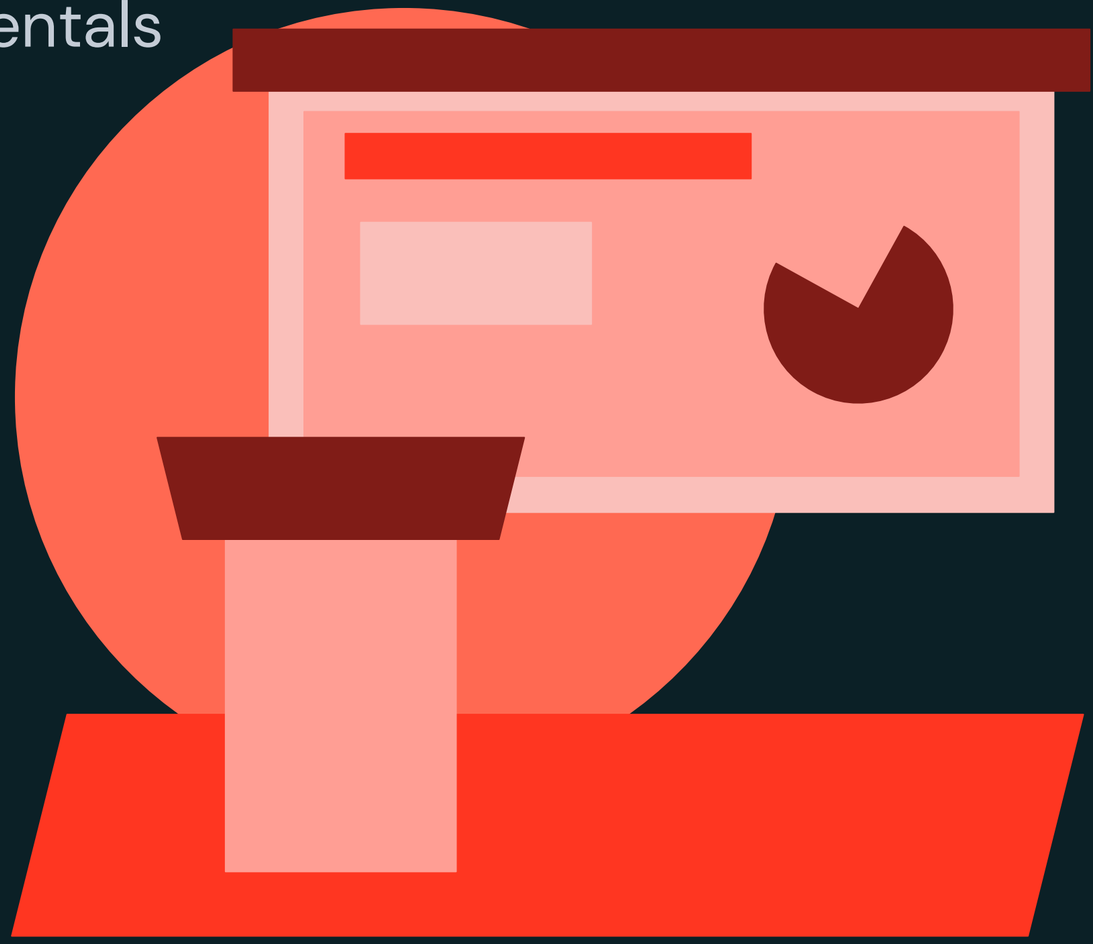
Notebook: 2 – Developing a Simple Pipeline

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

# Ensure Data Quality with Expectations



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Ensure Data Quality with Expectations

## Creating Expectations

With Lakeflow Spark Declarative Pipelines you can apply **data quality rules** to validate rows during ETL processes to gain insights into data integrity using the following

```
CONSTRAINT constraint-name EXPECT (column condition) [ON VIOLATION action]
```

| Action | SQL Syntax Example | Result |
|---|---|---|
| WARN (default) | `CONSTRAINT valid_notification EXPECT (notifications IN ('Y','N'))` | • Invalid rows **are still written** to the target<br>• Logs include **counts of valid vs. invalid records** and other metrics |
| DROP | `CONSTRAINT valid_date EXPECT (order_timestamp > "2021-01-01")`<br>`ON VIOLATION DROP ROW` | • Invalid rows **dropped** from table<br>• The count is logged alongside other metrics |
| FAIL | `CONSTRAINT valid_id EXPECT (customer_id IS NOT NULL)`<br>`ON VIOLATION FAIL UPDATE` | • Causes a **failure of a single flow** and does not cause other flows in your pipeline to fail.<br>• **Manual** intervention is required |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Ensure Data Quality with Expectations

## Adding Data Quality Expectations Example

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver
 (
   CONSTRAINT valid_notifications EXPECT (notifications IN ('Y','N')),
   CONSTRAINT valid_id EXPECT (customer_id IS NOT NULL) ON VIOLATION DROP ROW,
   CONSTRAINT valid_date EXPECT (order_timestamp > "2021-01-01") ON VIOLATION FAIL UPDATE
 )
AS
SELECT
      Order_id,
   timestamp(order_timestamp) AS order_timestamp,
   Customer_id,
   notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

WARN in metrics

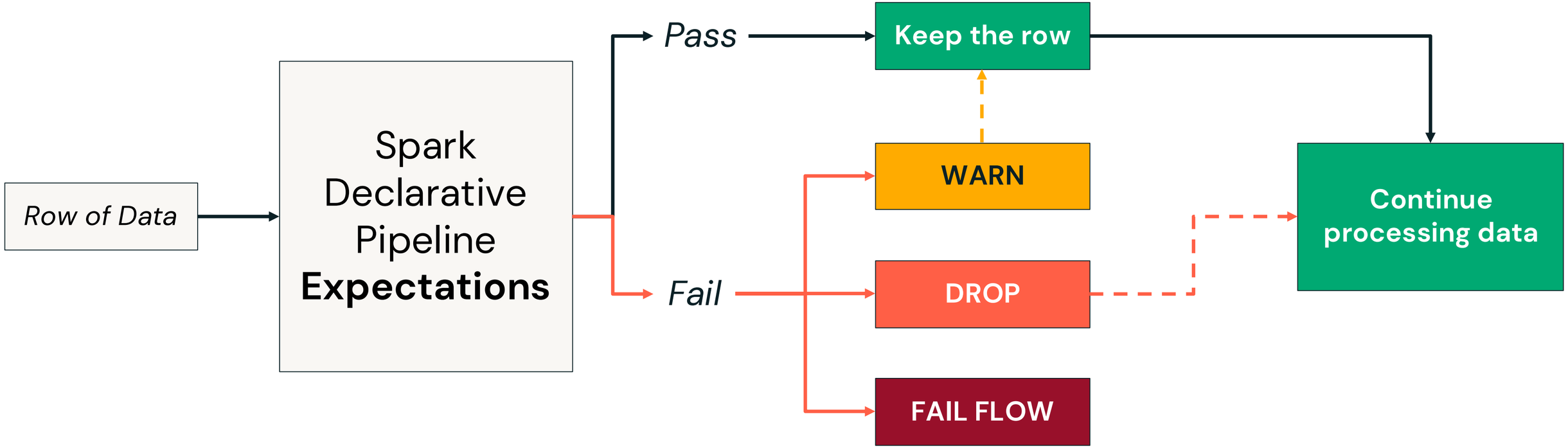
DROP the row

FAIL THE PIPELINE

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Ensure Data Quality with Expectations

## Actions Overview



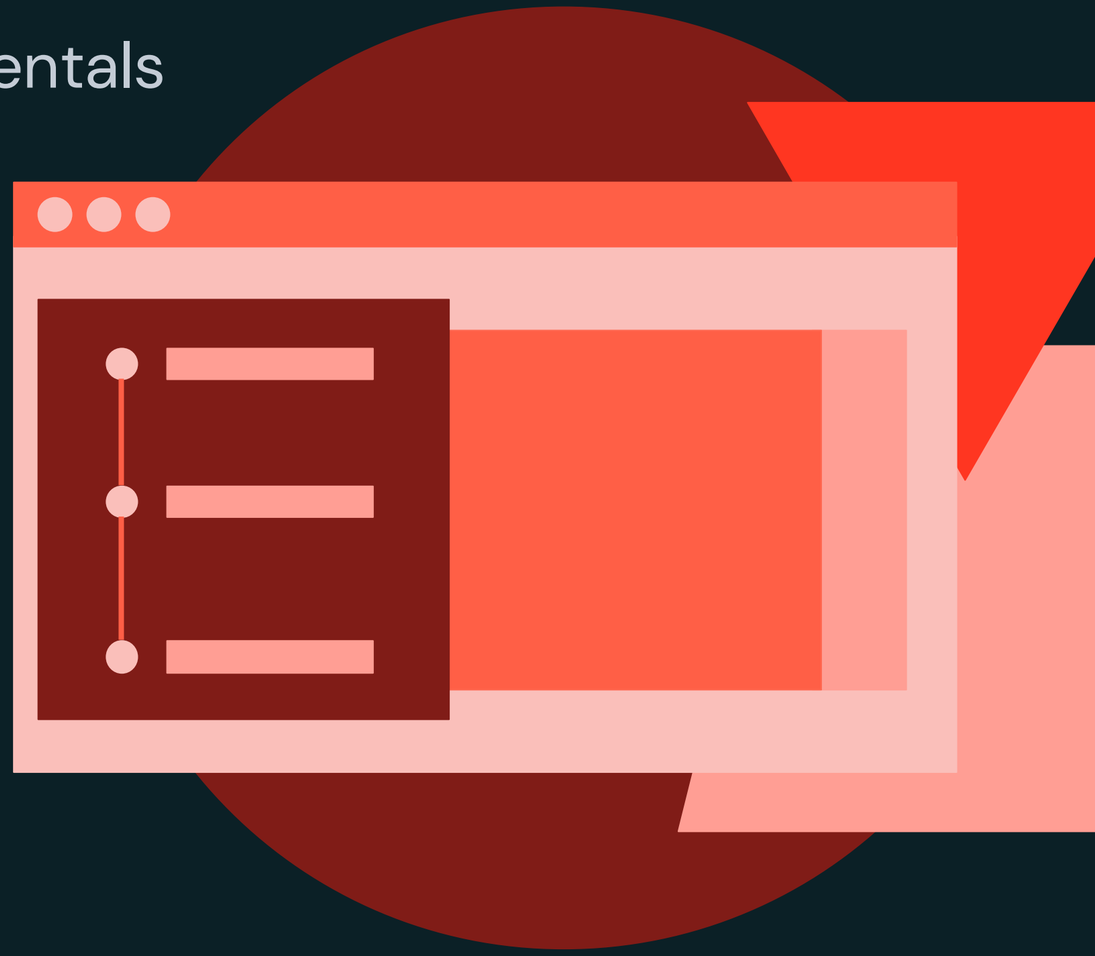

As of 2025Q2 materialized views that use **expectations** are **always fully refreshed**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

DEMONSTRATION

# Adding Data Quality Expectations



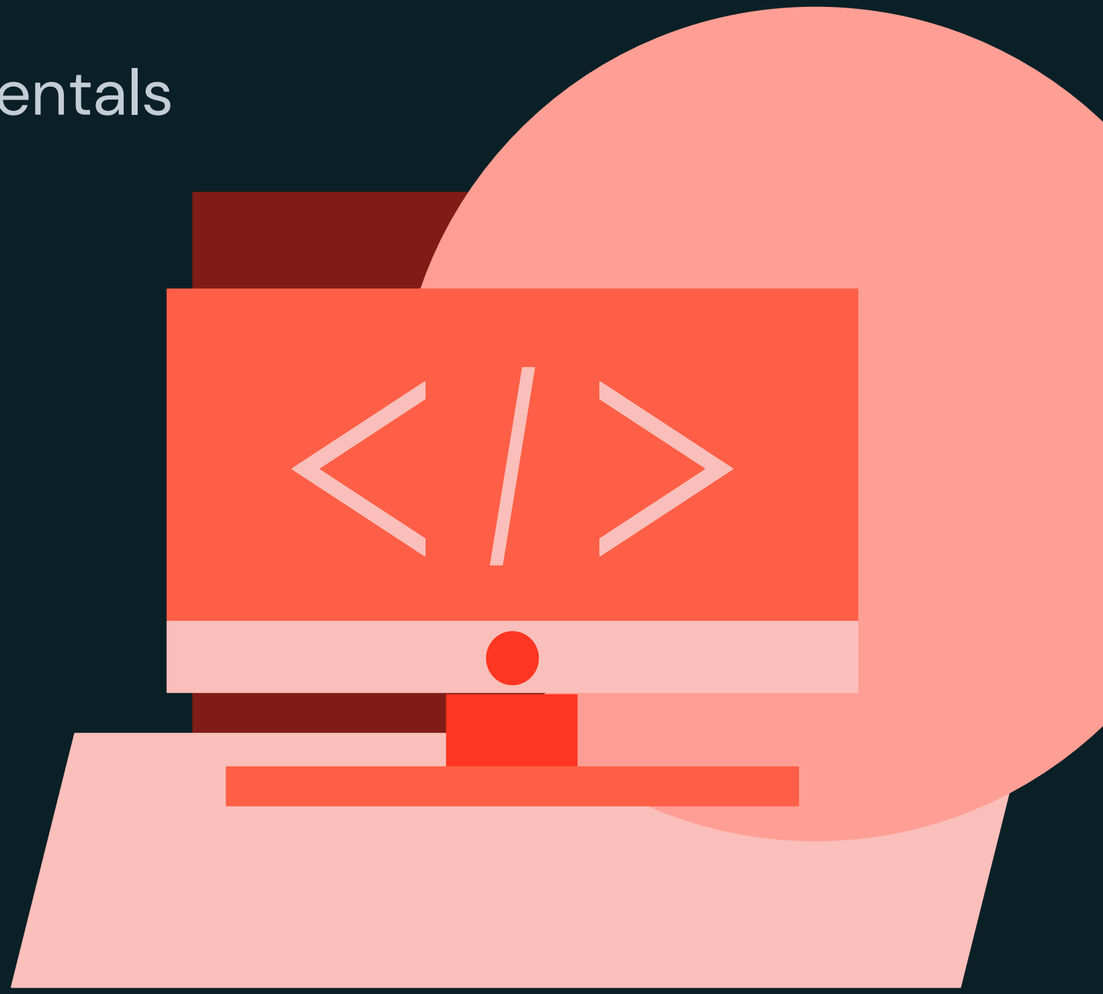
Notebook: 3 - Adding Data Quality Expectations

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Lakeflow Spark Declarative Pipeline Fundamentals

LAB EXERCISE

# Create a Pipeline



Notebook: 4 Lab - Create a Pipeline

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Building Lakeflow Spark Declarative Pipelines

**Build Data Pipelines with Lakeflow Spark Declarative Pipelines**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Section Learning Objectives

- Deploy a Lakeflow Spark Declarative Pipeline in production by modifying configuration options like mode, schedule, email notifications and more.
- Analyze event logs and pipeline metrics to examine the entirety of a pipeline.
- Design and implement a Change Data Capture (CDC) Declarative Pipeline using AUTO CDC INTO to handle slowly changing dimensions (SCD).

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Agenda

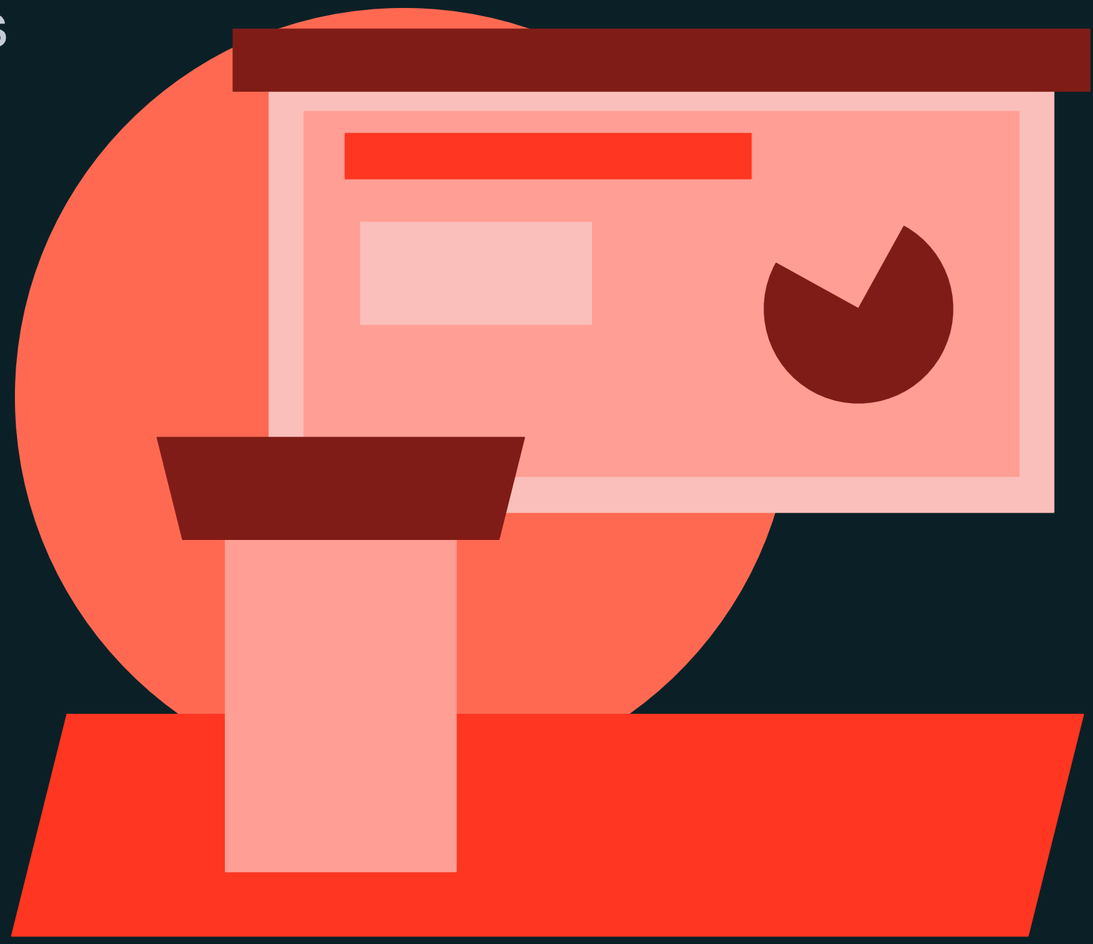
| Section Overview – Building Pipelines |
| --- |
| • Streaming Joins Overview |
| • Deploying a Pipeline to Production |
| • Change Data Capture (CDC) Overview |
| • Change Data Capture with AUTO CDC INTO |
| • Additional Features Overview |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

LECTURE

# Streaming Joins Overview



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Stream-Snapshot Join Overview

**Streaming Table** to **Static Table**

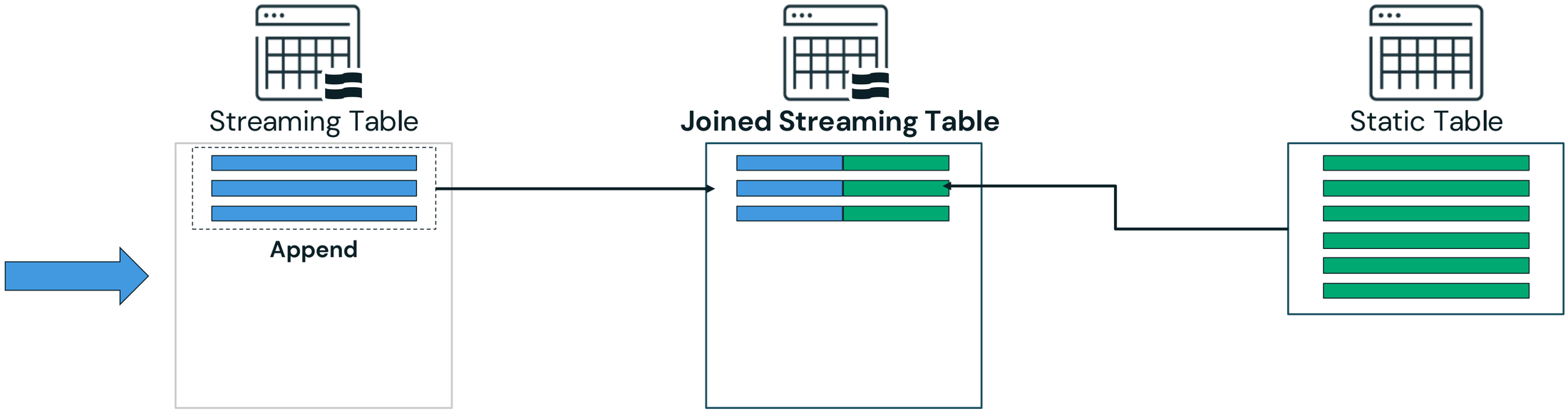
The goal is to **incrementally** join new data with a **static table**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Stream-Snapshot Join Overview

**Streaming Table** to **Static Table**



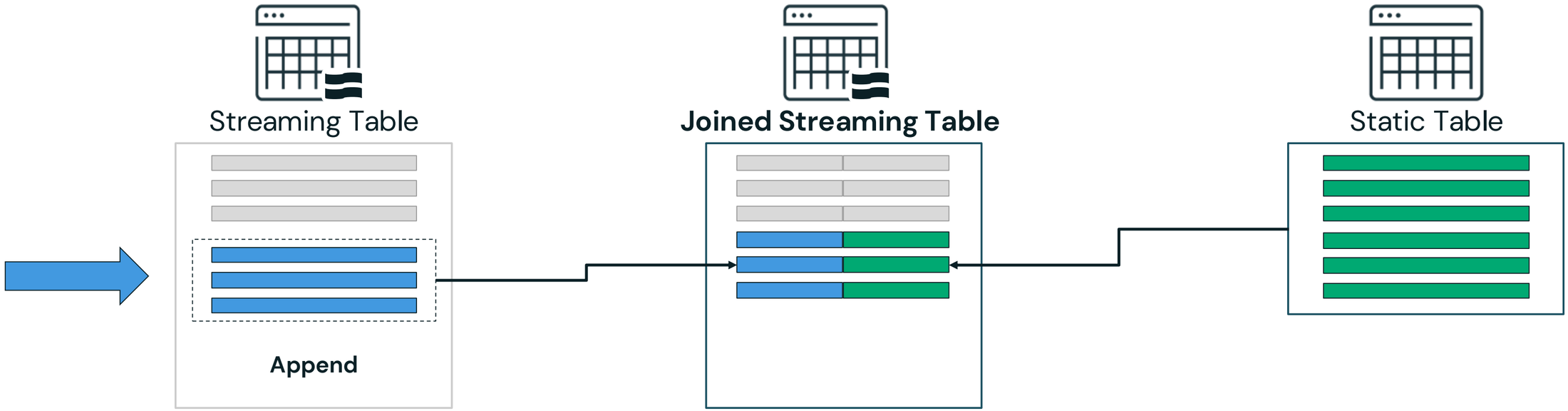

The **new data** in the streaming table JOINS using all of the **data in the lookup table**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Stream-Snapshot Join Overview

**Streaming Table** to **Static Table**




**Only the new data** in the streaming table JOINS using all of the **data in the lookup table**

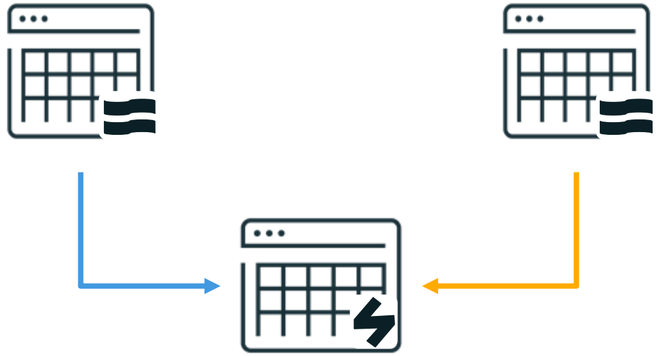
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

**Streaming Table** to **Streaming Table** with a **Materialized View**



The goal is to take **all rows from two streaming tables** and join them together each time the pipeline is run

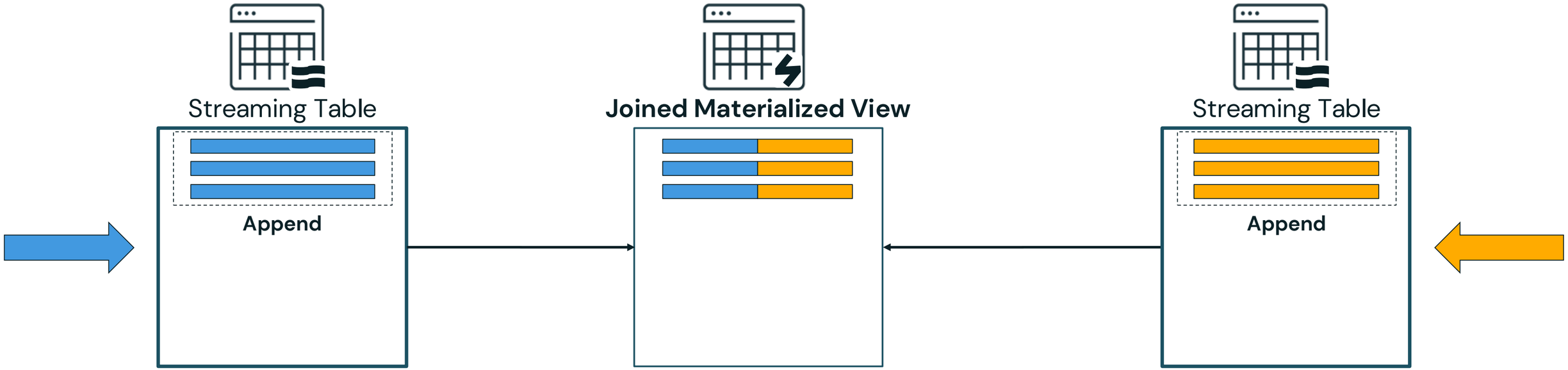
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

**Streaming Table** to **Streaming Table** with a **Materialized View**




The materialized view **efficiently computes all the data** in the streaming tables and joins the rows.

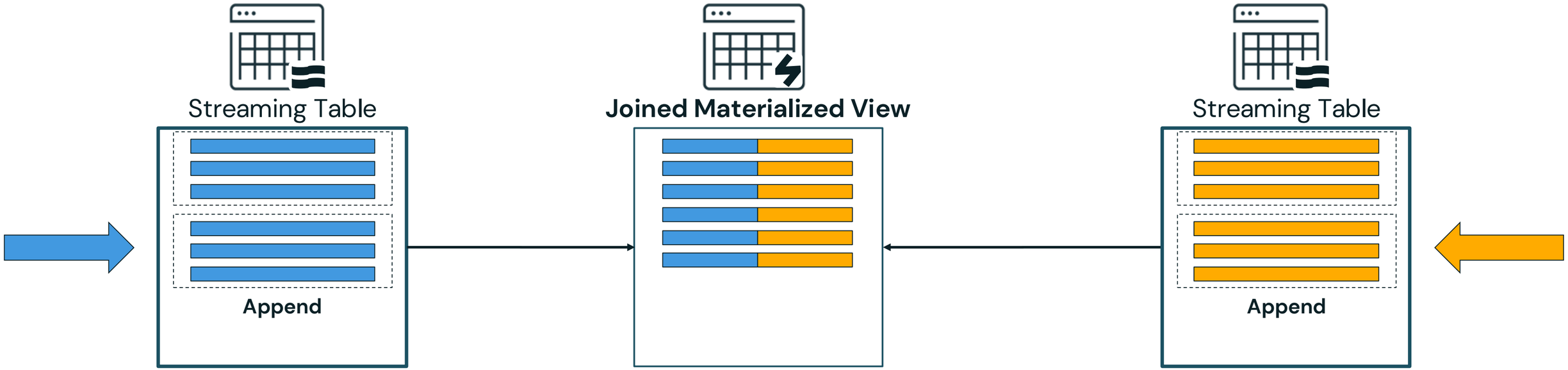
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

## Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

**Streaming Table** to **Streaming Table** with a **Materialized View**




**As new data is added** to the streaming tables, the materialized view will again **efficiently compute all the data** in the streaming tables and joins the rows

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Streaming Joins Overview

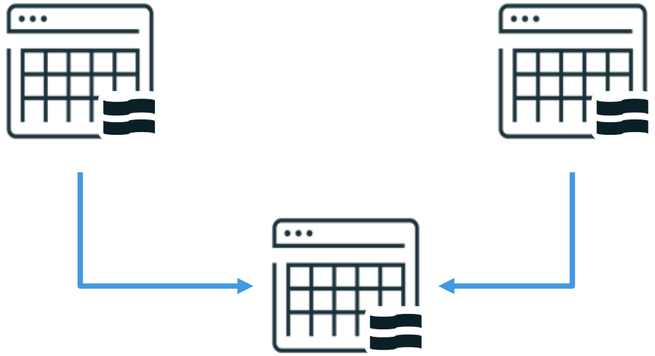
## Stream-Stream Joins

Outside the scope of the course

Streaming Table to Static Table

Streaming Table to Streaming Table with a Materialized View

**Incremental** Stream to Stream Joins



The goal is to **incrementally** join new data from two tables, **past data is not used**

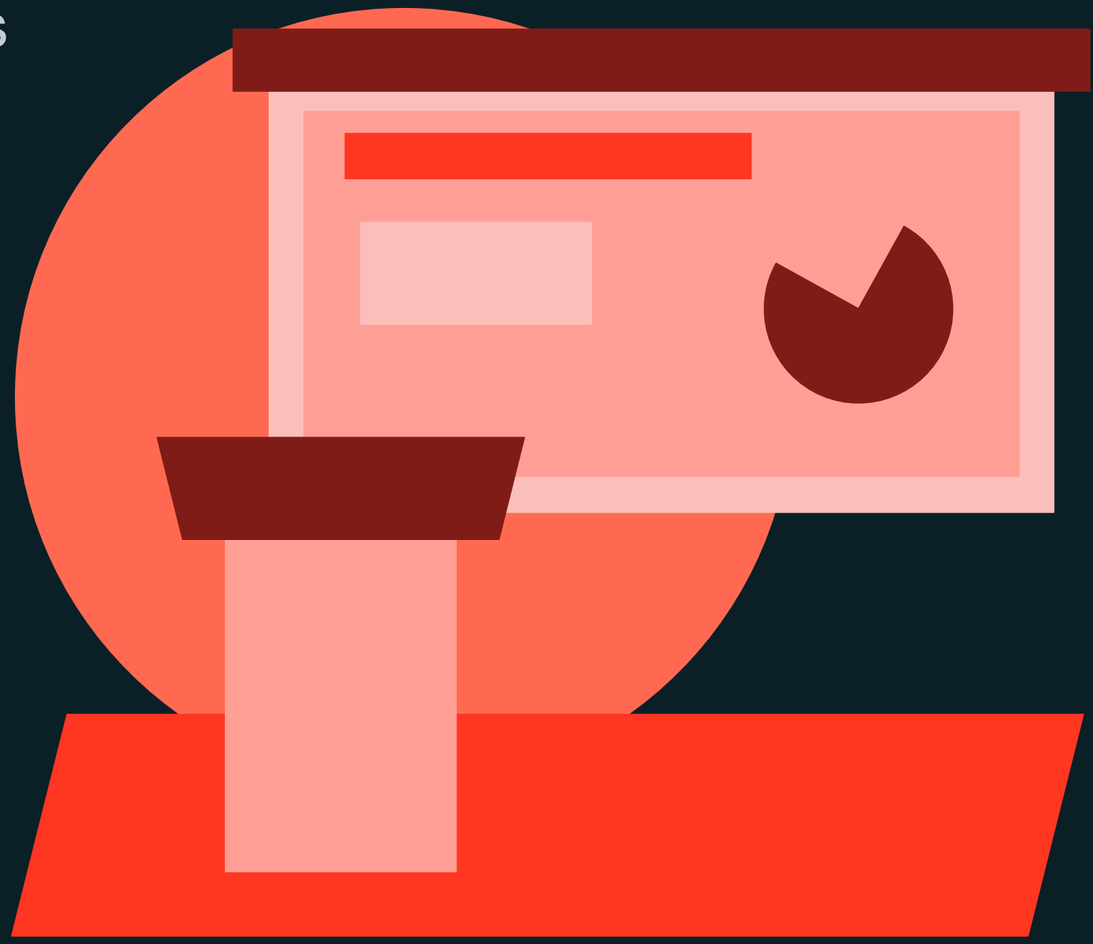
Databricks Streaming and DLT (Advanced) Course

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

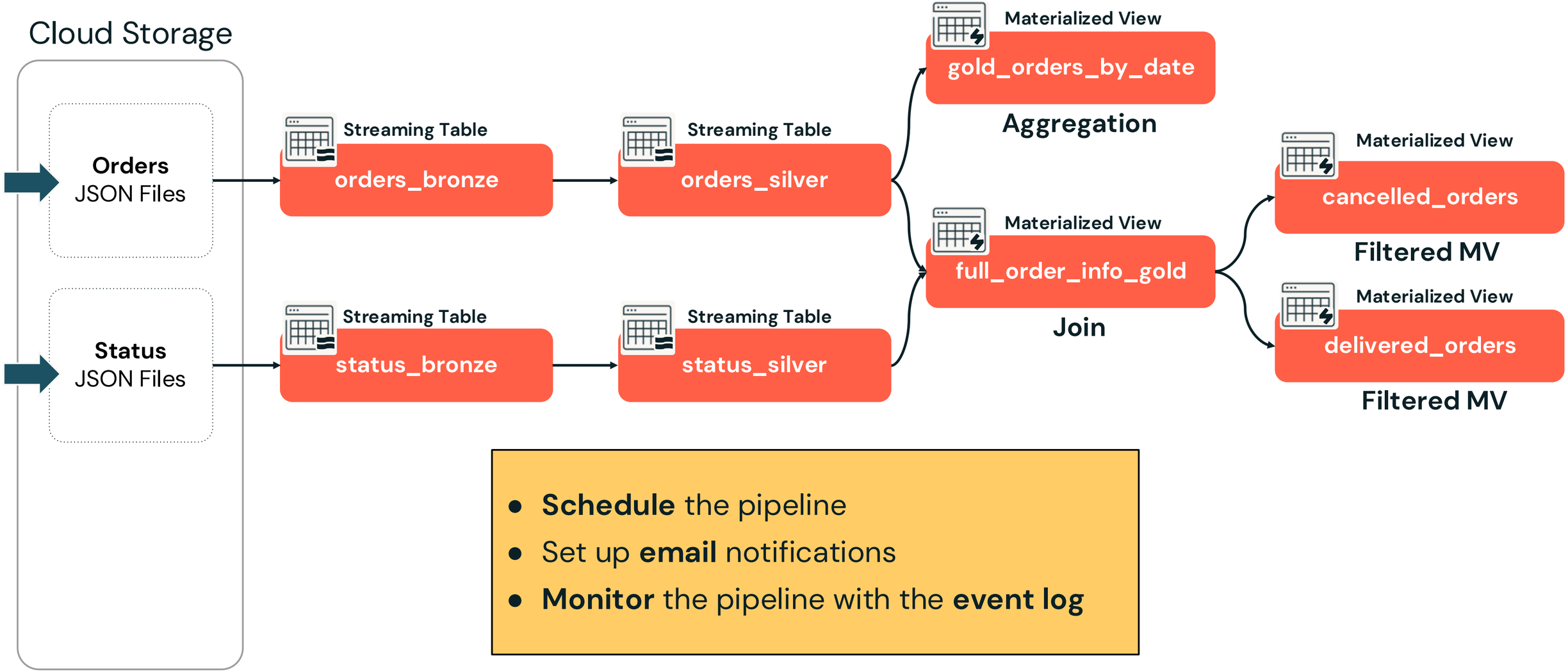
LECTURE

# Deploying a Pipeline to Production



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production




- **Schedule** the pipeline
- Set up **email** notifications
- **Monitor** the pipeline with the **event log**

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

## Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

## Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

### Triggered

- Can be triggered **manually** or run on a **schedule**
- **Refreshes** selected tables using data available at the start, then stops once the update is complete

### Continuous

- **Continuously** processes new data to keep streaming tables and materialized views up to date in **near real time**
- Monitors dependencies and **updates only** when source data changes

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

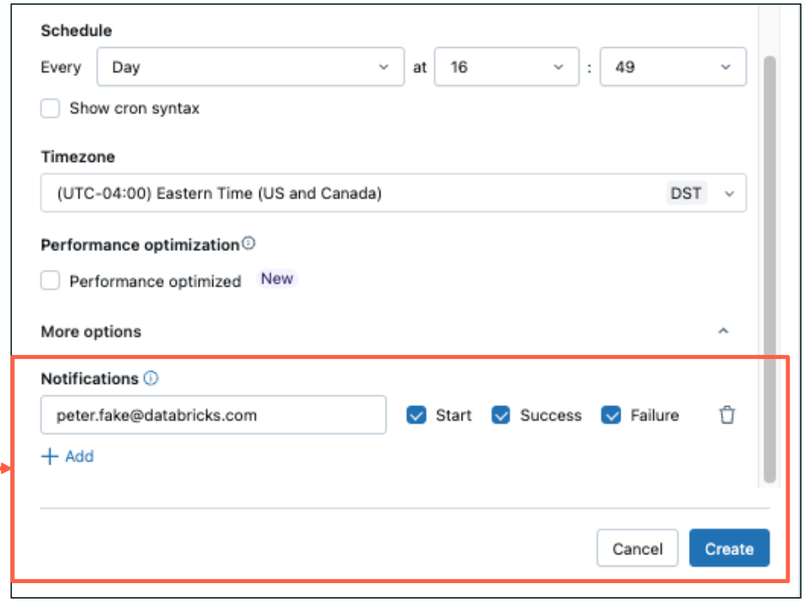
## Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

You can configure **email notifications** when **scheduling** the pipeline




© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

## Common Production Tasks

Scheduling

Email Notifications

Monitor pipelines

- The **pipeline event log** captures all key information about the pipeline:
  - Audit logs
  - Data Quality Checks
  - Pipeline Progress
  - Data Lineage

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

## Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

Query the pipeline event log table

```
SELECT * FROM <catalog>.<schema>.<event_log_table_name>
```

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Deploying a Pipeline to Production

## Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

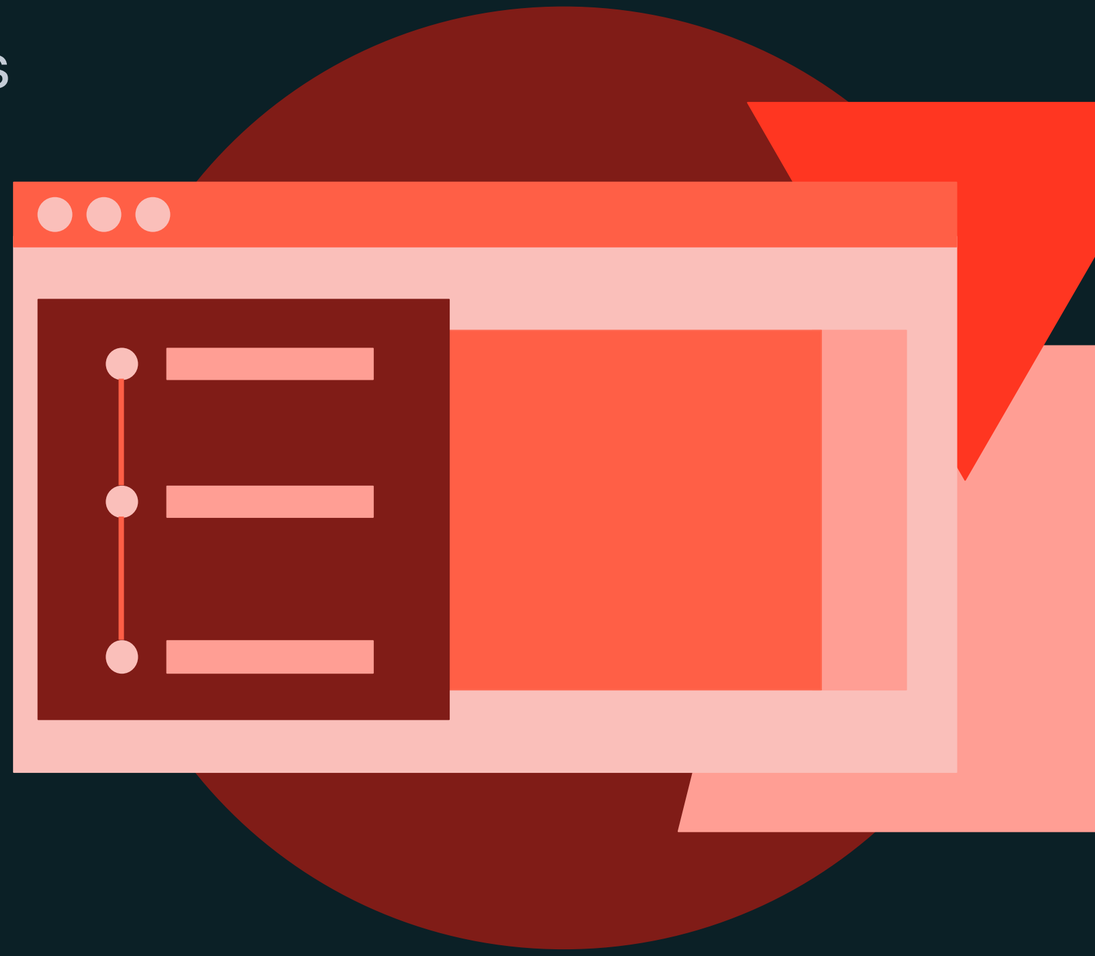
- **(DEFAULT)** The Event Log is written as a **hidden Delta table**
  - Located in the pipeline **default catalog** and **schema**
  - View the query the event log documentation to access the hidden event log

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

DEMONSTRATION

# Deploying Your Pipeline to Production



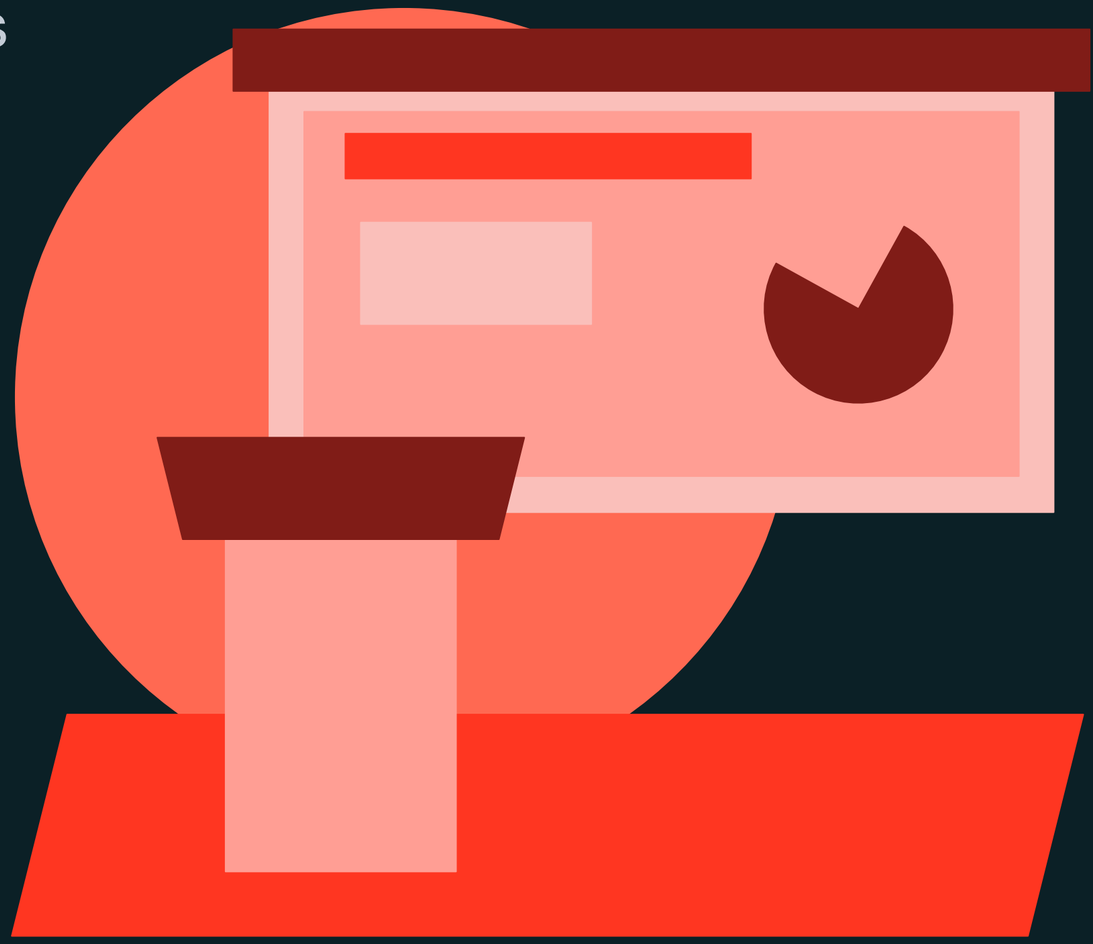
Notebook: 5 – Deploying Your Pipeline to Production

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

LECTURE

# Change Data Capture (CDC) Overview



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

## What is Change Data Capture?

- Change Data Capture (CDC) is a technique **used to track and capture changes in a data source** (such as a database, lakehouse or data warehouse).
- Then **applying those changes** into the Lakehouse or data warehouse

**Source Data**

INSERTS

UPDATES

DELETES

A Source table of changes, currently this must be streaming table

CDC is categorized into different types of **Slowly Changing Dimensions (SCD)**, which represent how **historical changes are handled**

Main types

**SCD Type 1**

**SCD Type 2**

Target Table

Table for changes to be applied

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

## Slowing Changing Dimensions (SCD) Type 1 (1 of 2)

### SCD Type 1 (Overwrite Target with Latest Values)

- When a record updates, the **previous record is simply overwritten** by its **key** with the new value.
- When a record is deleted by its **key**, the **record is removed**
- There is **no tracking of old keys (rows),** only the **current data** is retained.

customers (*target table with all customers*)

| CustomerID | Name | Address | ProcessDate |
|---|---|---|---|
| 1 | Peter | 1 Blue Rd. | 5/1/2025 |
| 2 | Samarth | 22 Front St | 5/1/2025 |

*Update the **customers** table with the **new customer** information*

updates (*source table with updates*)

| CustomerID | Name | Address | Operation | ProcessDate |
|---|---|---|---|---|
| 1 | Peter | **99 Sunny St..** | **UPDATE** | 5/15/2025 |
| 1 | Peter | **123 Main St.** | **UPDATE** | 5/20/2025 |
| 2 | null | null | **DELETE** | 5/20/2025 |
| 3 | Kostas | 100 Athens Dr. | **NEW** | 5/20/2025 |

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

## Slowing Changing Dimensions (SCD) Type 1 (2 of 2)

customers (*target table with all customers*)

Update the row

Delete the row

Insert the row

| CustomerID | Name | Address | ProcessDate |
|---|---|---|---|
| **1** | Peter | **1 Blue Rd.** | 5/1/2025 |
| **2** | Samarth | 22 Front St | 5/1/2025 |
| ***3*** | ***Kostas*** | ***100 Athens Dr.*** | ***5/20/2025*** |

updates (*source table with updates*)

| CustomerID | Name | Address | Operation | ProcessDate |
|---|---|---|---|---|
| 1 | Peter | **99 Sunny St..** | **UPDATE** | **5/15/2025** |
| 1 | Peter | **123 Main St.** | **UPDATE** | **5/20/2025** |
| 2 | null | null | **DELETE** | 5/20/2025 |
| 3 | Kostas | 100 Athens Dr. | **NEW** | 5/20/2025 |

### Updated SCD Type 1 Table

| CustomerID | Name | Address | ProcessDate |
|---|---|---|---|
| 1 | Peter | **123 Main St.** | 5/20/2025 |
| 3 | Kostas | 100 Athens Dr. | 5/20/2025 |

- **Peter** was updated with the **latest** information
- **Samarth** was deleted from the table
- **Kostas** was added to the table

Used **SCD Type 1** when only the **current data matters** and historical values are irrelevant

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

## Slowing Changing Dimensions (SCD) Type 2 (*Outside the scope of the course*)

### SCD Type 2 (Historical Tracking/Versioning)

- The **old record is preserved** with an additional column indicating its validity period (start date, end date, or a current flag) and a **new record is inserted** with the updated information.
- When a **record is marked as deleted,** the record is kept and a column indicates the record is **inactive**.

- Used when **historical data is important**, and the system needs to track **how attributes change over time** (like tracking changes in a customer's address or status).

Columns indicate **active** and **inactive** rows

**Example customers target table with SCD Type 2**

| CustomerID | Name | Address | ProcessDate | __START_AT | __END_AT |
|---|---|---|---|---|---|
| 1 | Peter | 123 Main St. | 5/20/2025 | 5/20/2025 | null |
| 1 | Peter | 1 Blue Rd. | 5/1/2025 | 5/1/2025 | 5/20/2025 |
| 2 | Samarth | 22 Front St | 5/1/2025 | 5/1/2025 | 5/20/2025 |
| 3 | Kostas | 100 Athens Dr. | 5/20/2025 | 5/20/2025 | null |

**Null** values indicate the **current row**

**Non null** values indicate **inactive (historic)** rows

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

## Using AUTO CDC INTO in Spark Declarative Pipelines (Formerly APPLY CHANGES INTO)

customers (*target table with all customers*)

| CustomerID | Name | Address | ProcessDate |
|---|---|---|---|
| 1 | Peter | 1 Blue Rd. | 5/1/2025 |
| 2 | Samarth | 22 Front St | 5/1/2025 |

updates (source table with updates)

| CustomerID | Name | Address | Operation | ProcessDate |
|---|---|---|---|---|
| 1 | Peter | 99 Sunny St.. | **UPDATE** | 5/15/2025 |
| 1 | Peter | 123 Main St. | **UPDATE** | 5/20/2025 |
| 2 | Samarth | 22 Front St | **DELETE** | 5/20/2025 |
| 3 | Kostas | 100 Athens Dr. | **NEW** | 5/20/2025 |

```
CREATE OR REFRESH STREAMING TABLE customers;

CREATE FLOW scd_type_1_flow AS
AUTO CDC INTO customers
 FROM STREAM updates
 KEYS (CustomerID)
 APPLY AS DELETE WHEN operation = "DELETE"
 SEQUENCE BY ProcessDate
 COLUMNS * EXCEPT (operation)
 STORED AS SCD TYPE 1;
```

- Create a streaming table, then use AUTO CDC to populate it
- Apply updates, inserts and deletes to the **target** table
- Source records to determine **updates, deletes and inserts**
- The column(s) that **uniquely identify a row** in the source and target tables
- Specifies when a CDC event should be treated as a **DELETE** rather than an upsert.
- Specifies the **logical order of CDC events** in the source data
- Specifies a **subset of columns** to include in the **target**
- Whether to store records as **SCD type 1 (default) or type 2**

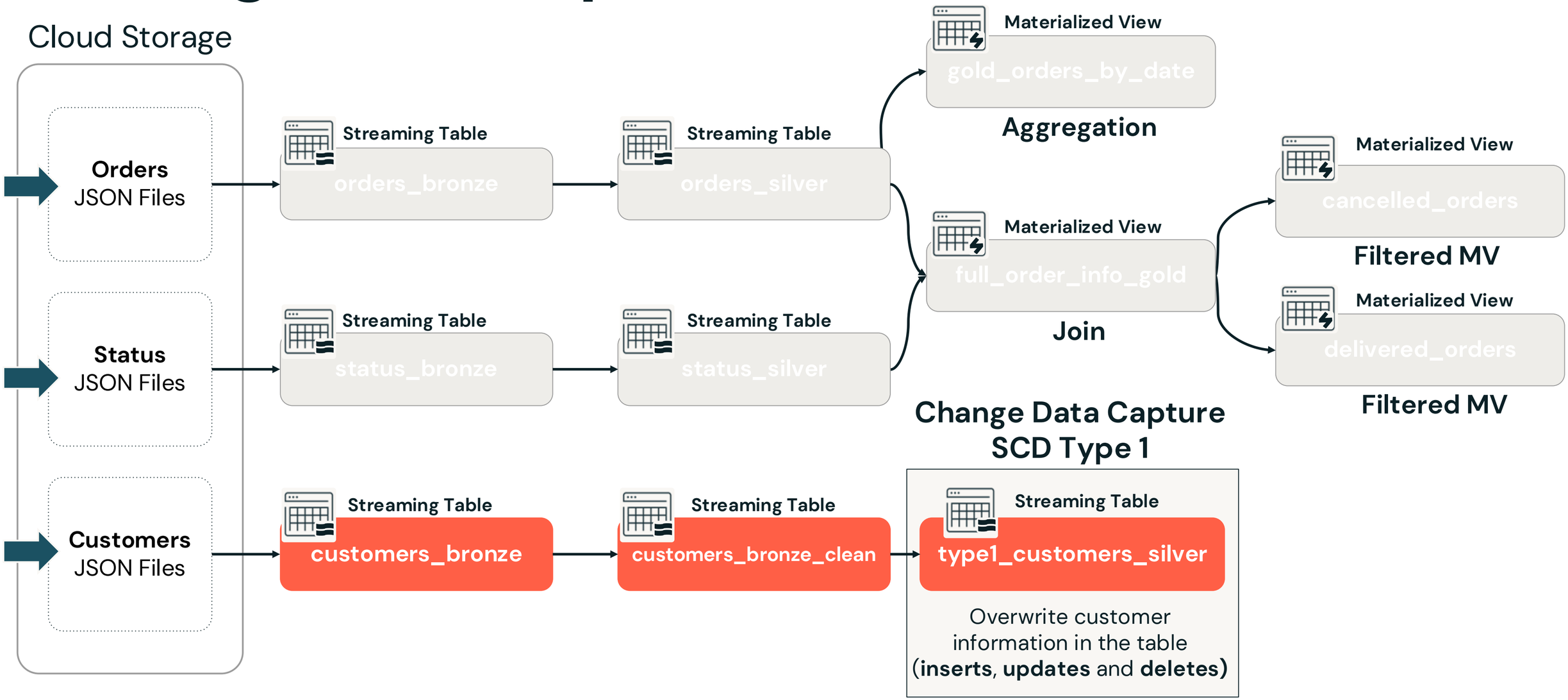
*INSERTs and UPDATEs are implicitly implemented using the **KEY(s)**, no coding required*

Use **`AUTO CDC INTO (Formerly APPLY CHANGE INTO)`** instead of a traditional batch job with complex **`MERGE INTO`** `syntax`

Use `AUTO CDC FROM SNAPSHOT` (Public Preview as of 2025Q2) to process changes in database snapshots.

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Change Data Capture (CDC) Overview

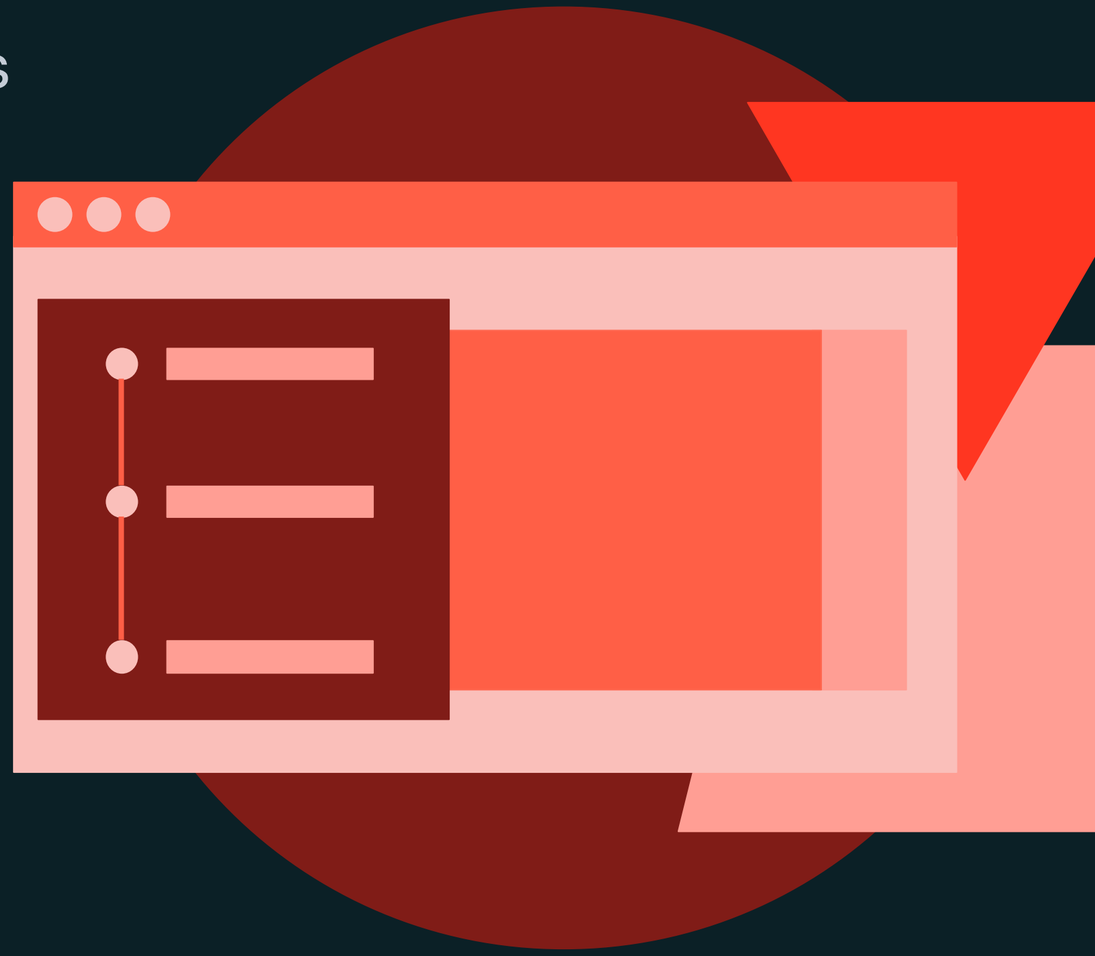



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

DEMONSTRATION

# Change Data Capture with AUTO CDC INTO INTO



Notebook: 6 – Change Data Capture with AUTO CDC with SCD TYPE 1

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

LAB EXERCISE

# BONUS - AUTO CDC INTO with SCD Type 1

**If you are interested in performing CDC with Spark Declarative Pipelines we have provided an extra lab you can complete after class.**

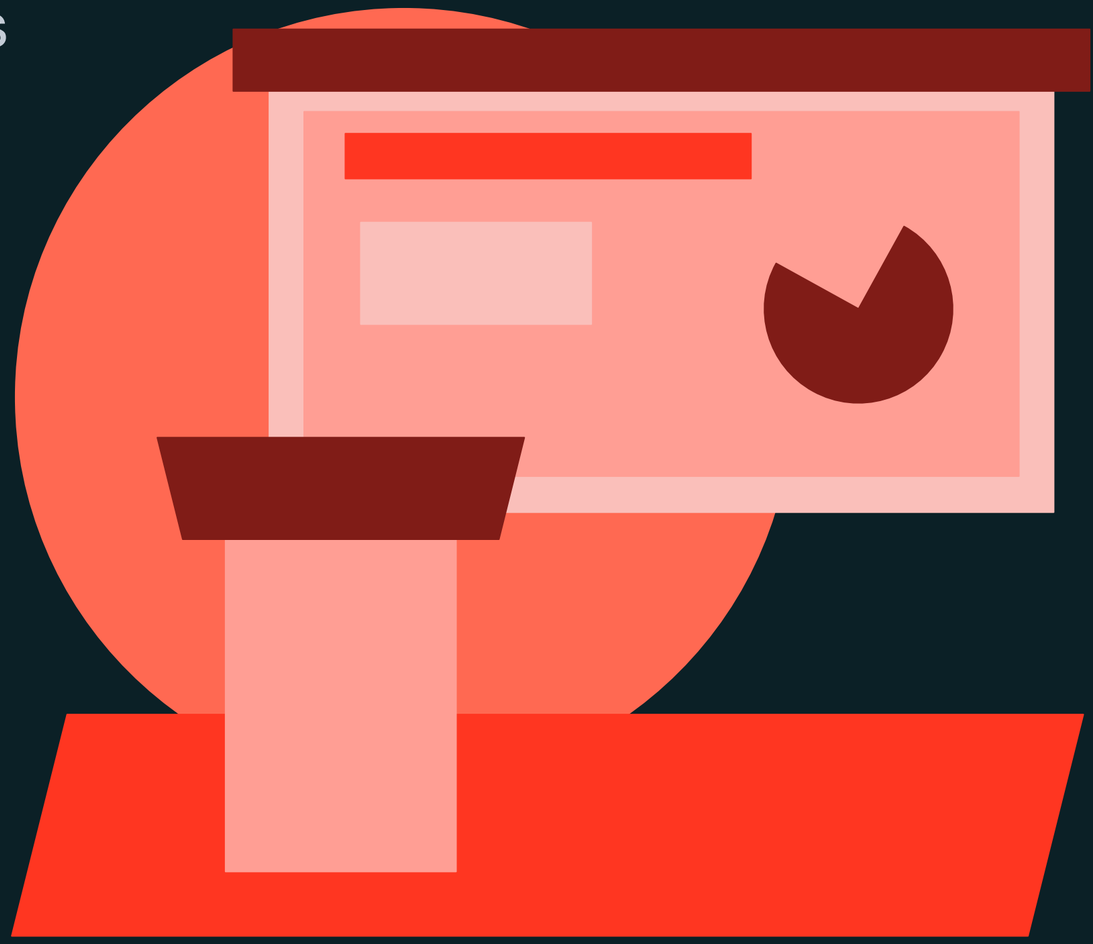
Notebook: 7 BONUS Lab - AUTO CDC INTO with SCD Type 1

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

Building Lakeflow Spark Declarative Pipelines

LECTURE

# Additional Features Overview



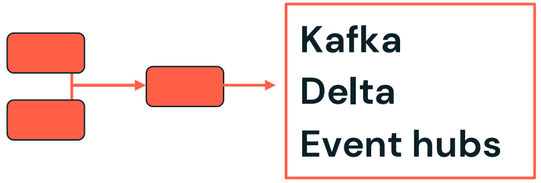
© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Additional Features Overview

## What's Next: Features Outside This Course

### FLOWS

Add **flows** for the **same target**, creating a union of the two data sources (or tables)




### SINKS

Sink API **enables writing to destinations** outside of the pipeline:

- Kafka
- Delta
- Event hubs

### Full DELTA Support

Improved Delta Support for MVs/STs

- Liquid clustering
- Row-level security and column masking
- Change data feed (CDF) from STs

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Additional Features Overview

## What's Next: Features Outside This Course

### Full UC Support

- Publish to multiple catalogs & schemas
- Read **STs** and **MVs** in Dedicated Access Mode
- Create a Unity Catalog pipeline by cloning a Hive metastore pipeline

### Better Performance

Improved performance throughout Lakeflow Spark Declarative Pipelines!

- Serverless
- Incremental refresh for materialized views
- Photon

### Databricks Asset Bundles (DABs)

DABs enable you to **programmatically** validate, deploy, and run Databricks resources such as pipelines for **CI/CD production Lakeflow Jobs**

- Automated Deployment with Databricks Asset Bundles course

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Summary and Next Steps

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Sp
Iceberg logo are trademarks of the Apache Software Foundation.

# Course Learning Objective Recap

- Understand the core concepts and components of Lakeflow Spark Declarative Pipelines, including the function and differences between streaming tables, materialized views, and temporary views.
- Identify and configure Lakeflow pipeline settings, such as compute, data assets, trigger modes, and advanced options.
- Develop a functional Lakeflow Spark Declarative Pipeline using the new pipeline editor and SQL-based syntax.
- Incorporate data quality expectations into a pipeline to validate and enforce data integrity.
- Analyze event logs and pipeline metrics to understand the full execution and lifecycle of a Lakeflow Spark Declarative Pipeline.
- Design and implement a Change Data Capture (CDC) to a pipeline using AUTO CDC INTO to handle slowly changing dimensions (SCD).

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

# Next Steps

## Additional resources for continuing the learning journey.

### Data Engineering with Databricks

- Continue your learning through self-paced or instructor-led offerings
- The courses offer hands-on instruction in:
  - Databricks Data Science & Engineering Workspace
  - Databricks SQL
  - Spark Declarative Pipelines
  - Databricks Repos
  - Databricks Task Orchestration
  - Unity Catalog

### Data Engineer Associate Certification

- Validate your data and AI skills on Databricks by earning a Databricks credential
- Exam information and exam guide
- The exam covers:
  - Data Intelligence Platform
  - ELT With Spark SQL and Python
  - Incremental Data Processing
  - Production Pipelines
  - Data Governance

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

databricks

© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.